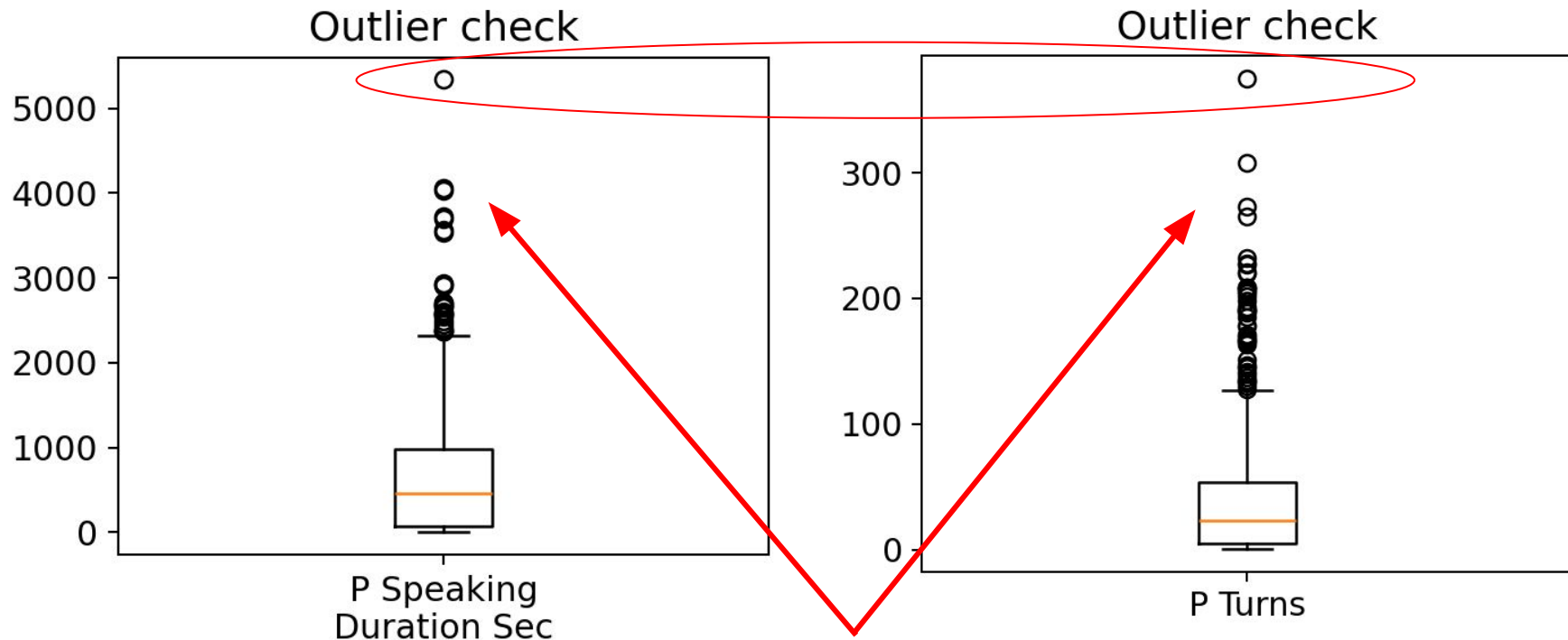


Max Progress

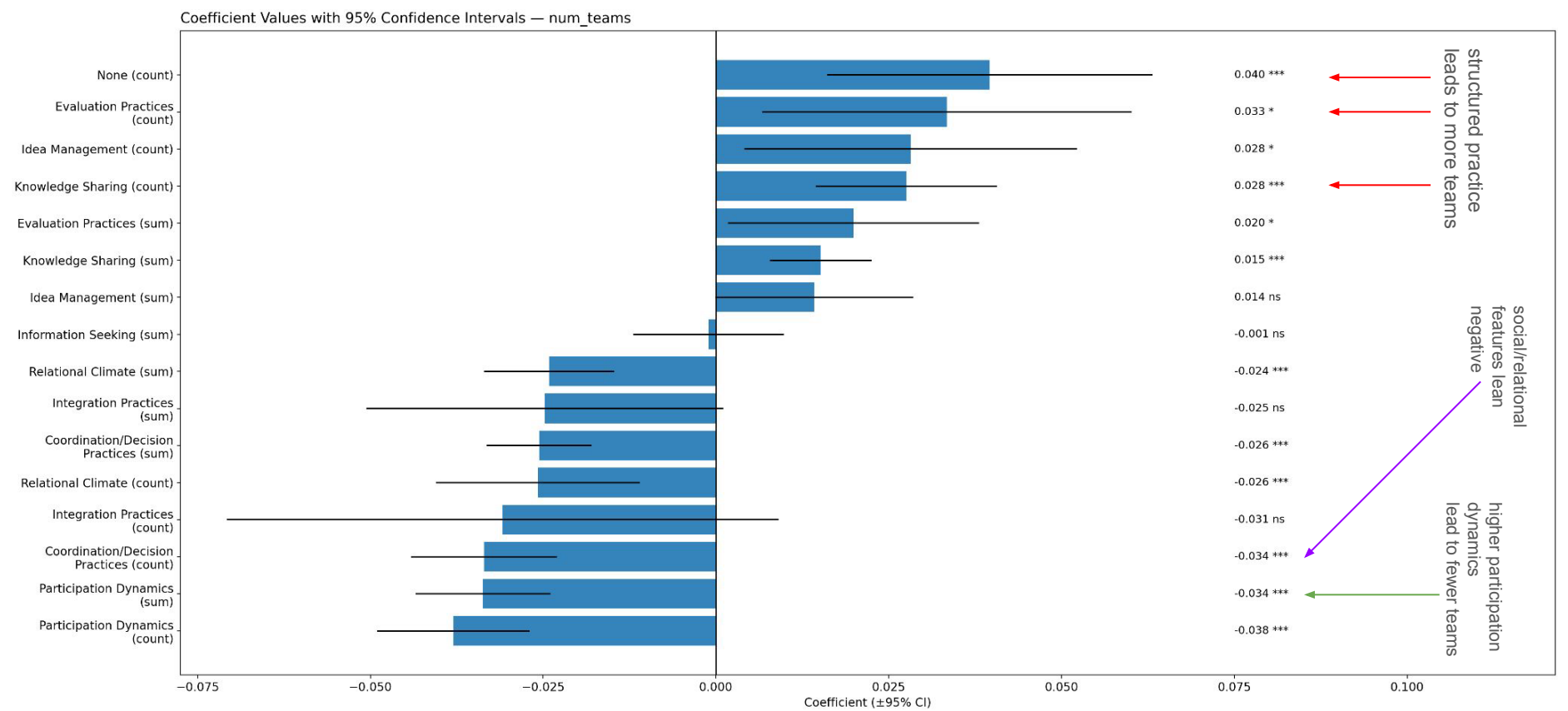
Description

- What questions are we trying to answer?
 - Are there differences between people who are and are not facilitators / in a team/ in a funded team?
 - Temporal: nuanced differences that may occur at different timepoints of a meeting
- What are the features? (across first, middle, and end)
 - What strategies are you using: Sum, count
 - What is the quality of the strategies: Average score for each strategy (in first , middle, and end)
- What does it mean to aggregate across persons instead of sessions? Show how you re aggregating with a simple drawing or something
- What regression model you used and why?
 - Control variables and why?
 - Outcome? (num_teams, num_funded_teams (distribution of these variables); if a lot of zeros in outcomes, maybe worth looking into neg binomial
 - Logistic regression (binary outcome : facilitator or not?)
- Findings:
 - People who do xxx at the (what time) of a meeting is more likely to be a facilitator/ in a team/ or in a funded team
 - What could this mean?

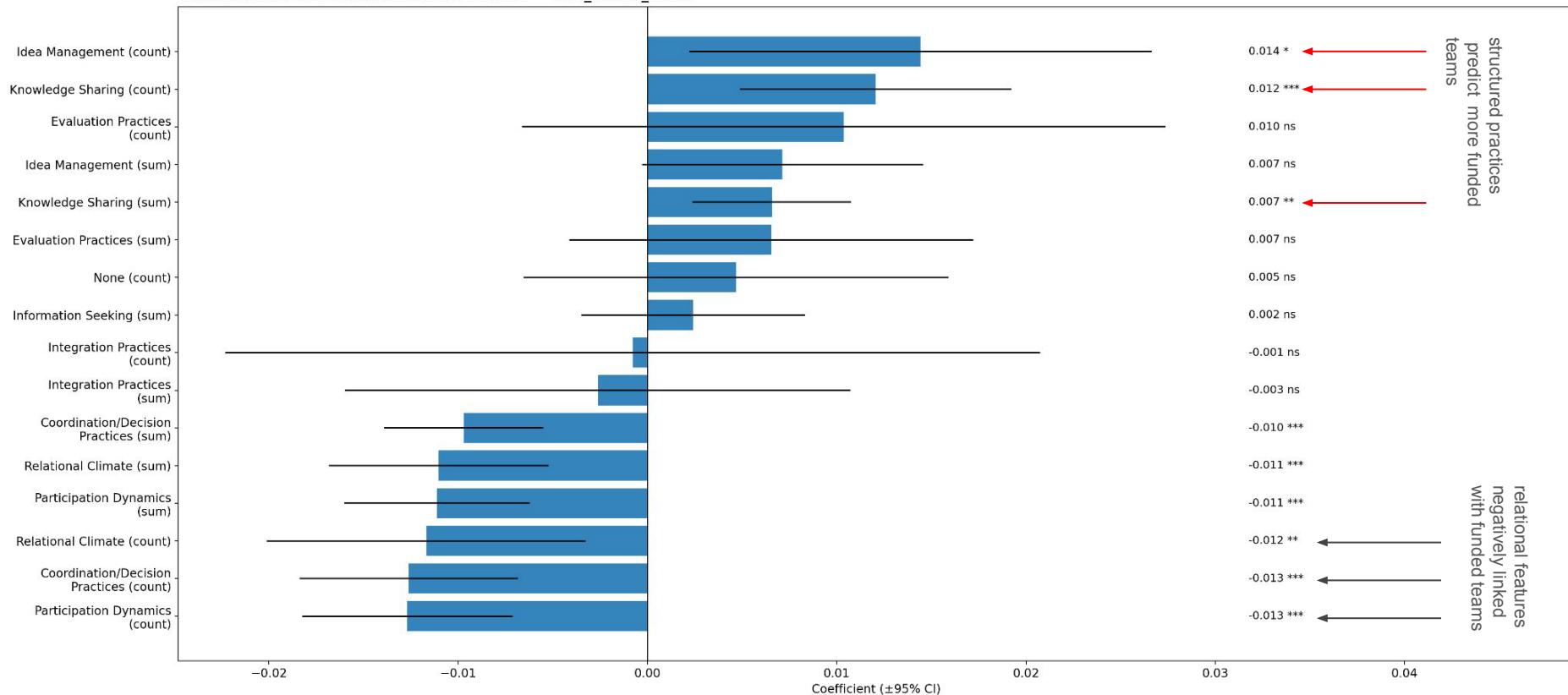


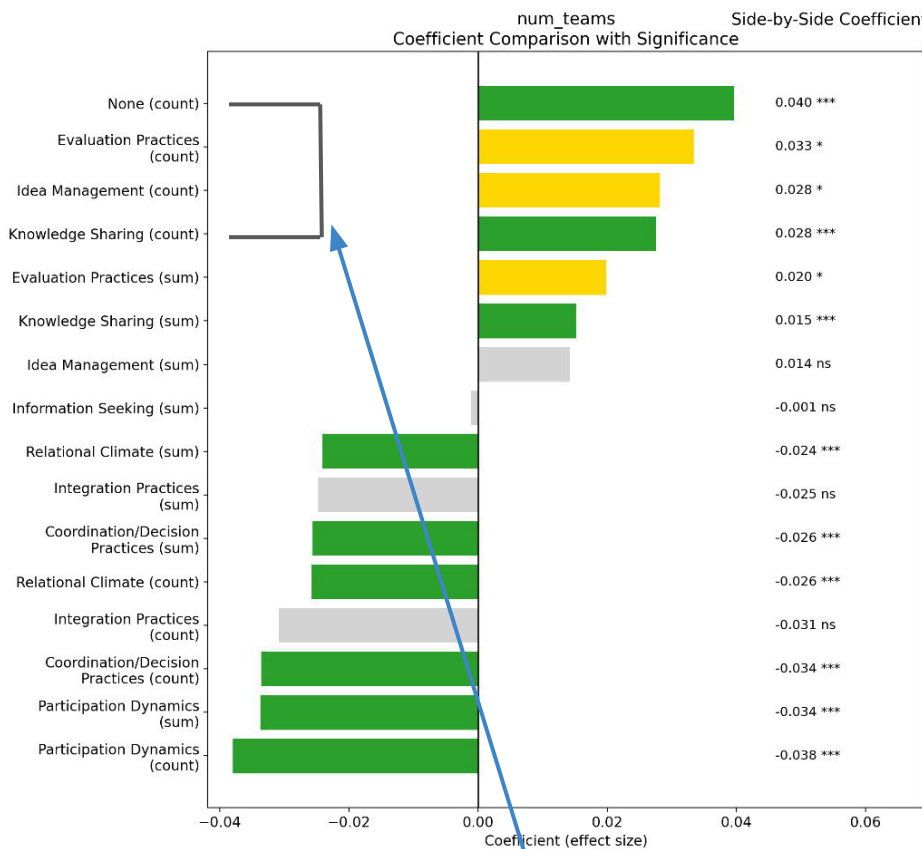
Both speaking duration & number of turns
show long right tails trimmed before regression analysis

Linear regression: control?

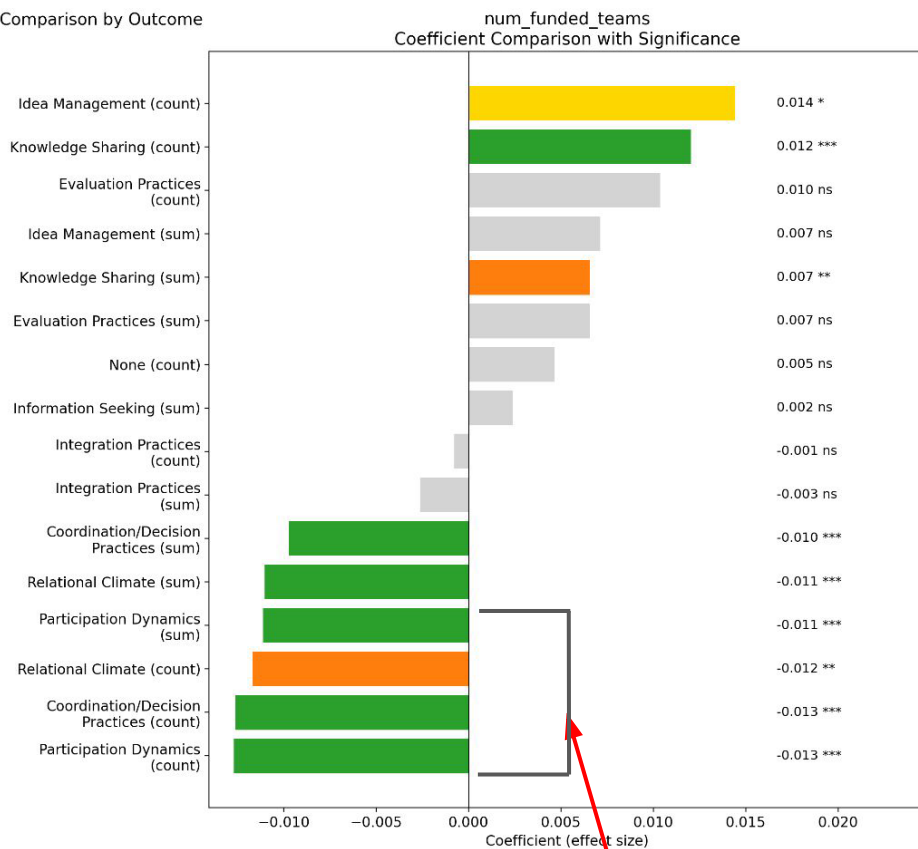


Coefficient Values with 95% Confidence Intervals — num_funded_teams

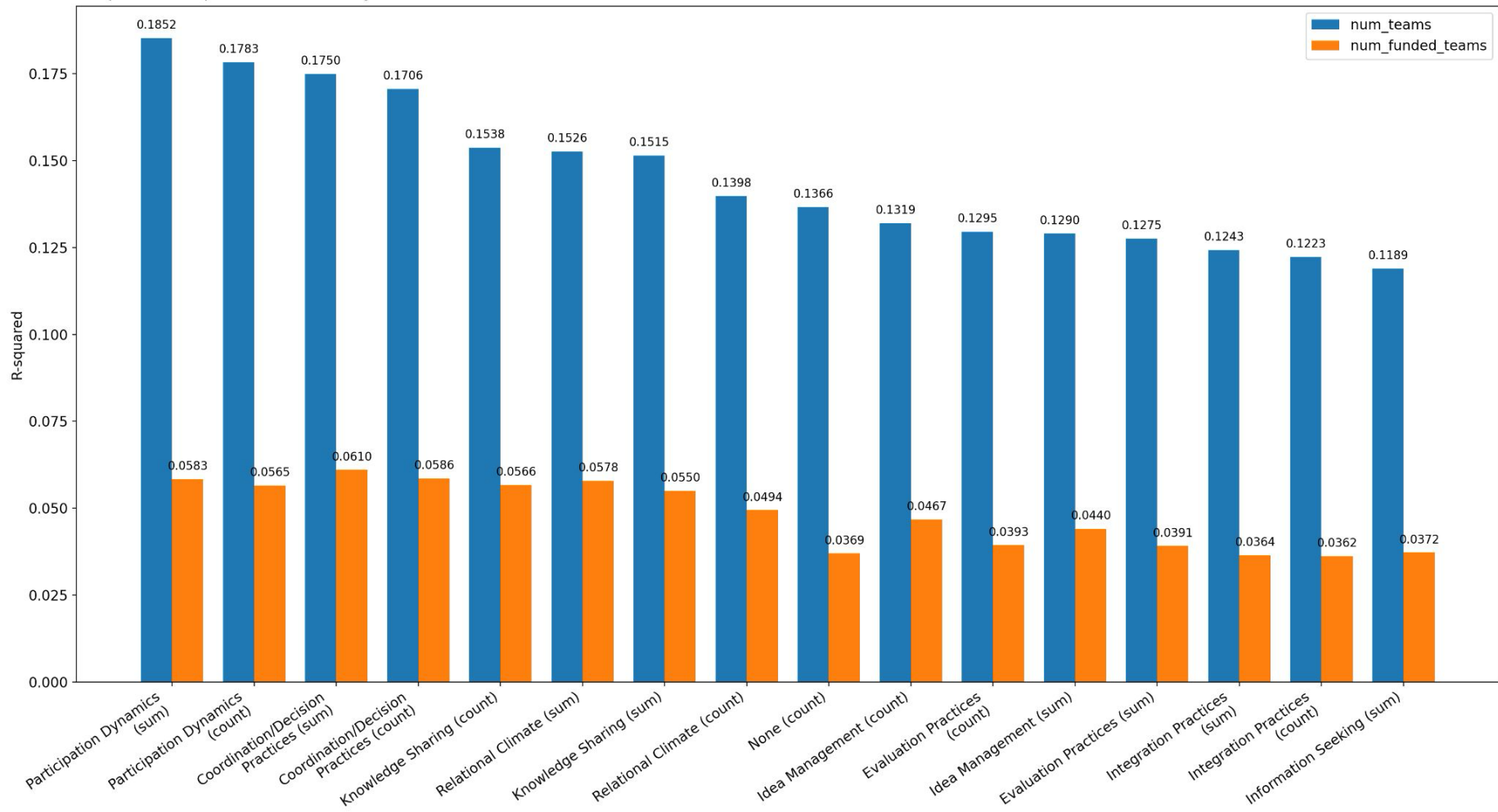




More & stronger predictors for team count



Participation dynamics consistently negative

R-squared Comparison — Sorted by Maximum R²

Significant Results Only — *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Participation Dynamics (sum)	***	***
Participation Dynamics (count)	***	***
Coordination/Decision Practices (sum)	***	***
Coordination/Decision Practices (count)	***	***
Knowledge Sharing (count)	***	***
Relational Climate (sum)	***	***
Knowledge Sharing (sum)	***	**
Relational Climate (count)	***	**
None (count)	***	ns
Idea Management (count)	*	*
Evaluation Practices (count)	*	ns
Idea Management (sum)	ns	ns
Evaluation Practices (sum)	*	ns
Integration Practices (sum)	ns	ns
Integration Practices (count)	ns	ns
Information Seeking (sum)	ns	ns

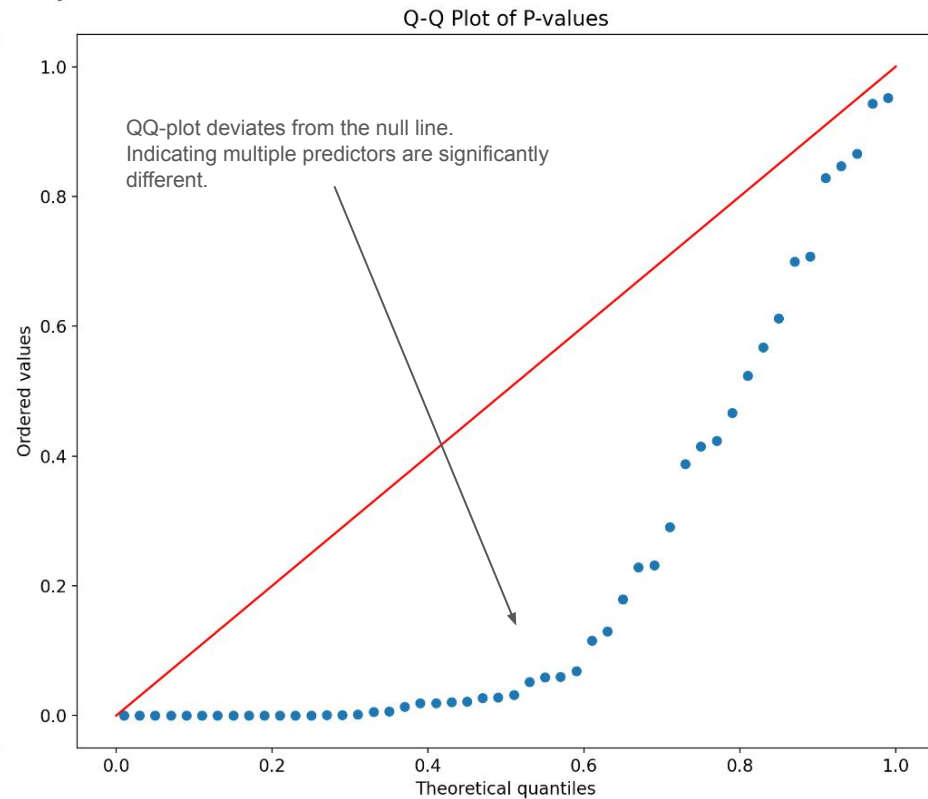
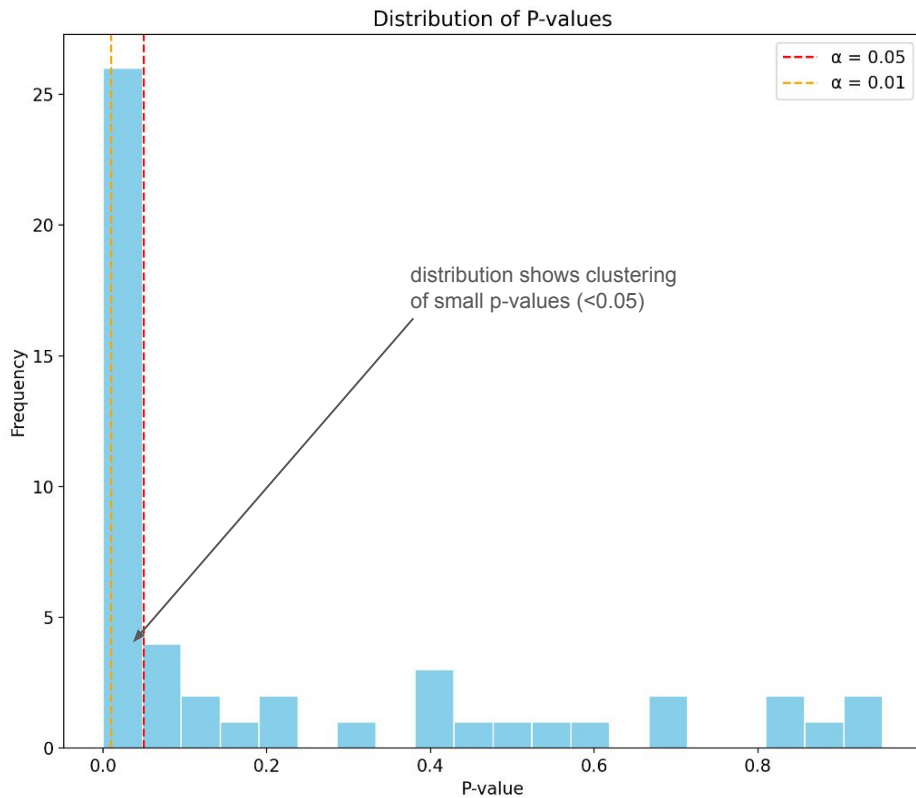
num_teams

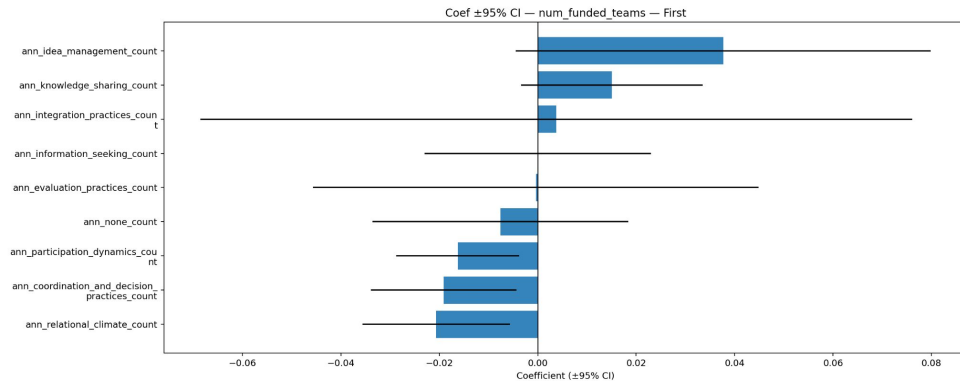
num_funded_teams

Strong predictors for team count

Few predictors significant for funded teams

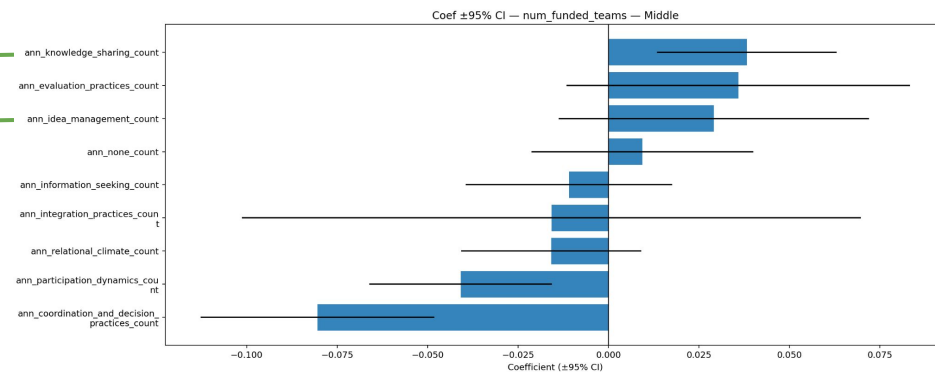
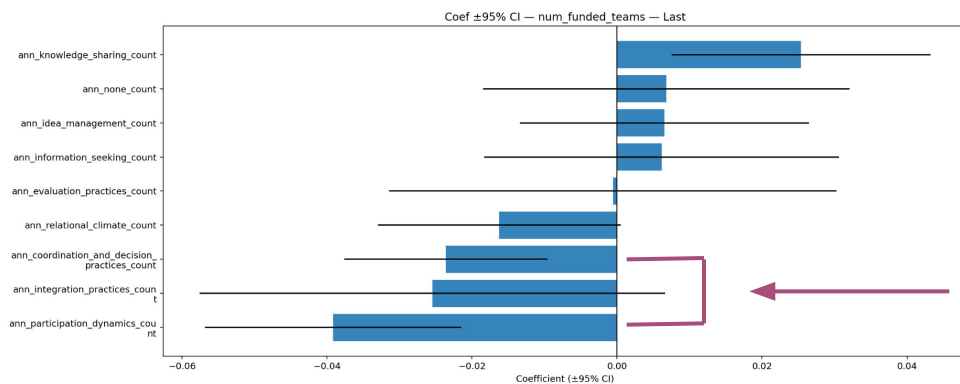
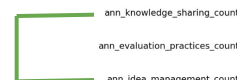
P-value Analysis





Idea management and knowledge-sharing behaviors are more predictive of team outcomes - reflecting active idea exchange and framing of course of conversation for session.

Coordination and integration behaviors become more predictive, indicating a shift from exploration to structured collaboration



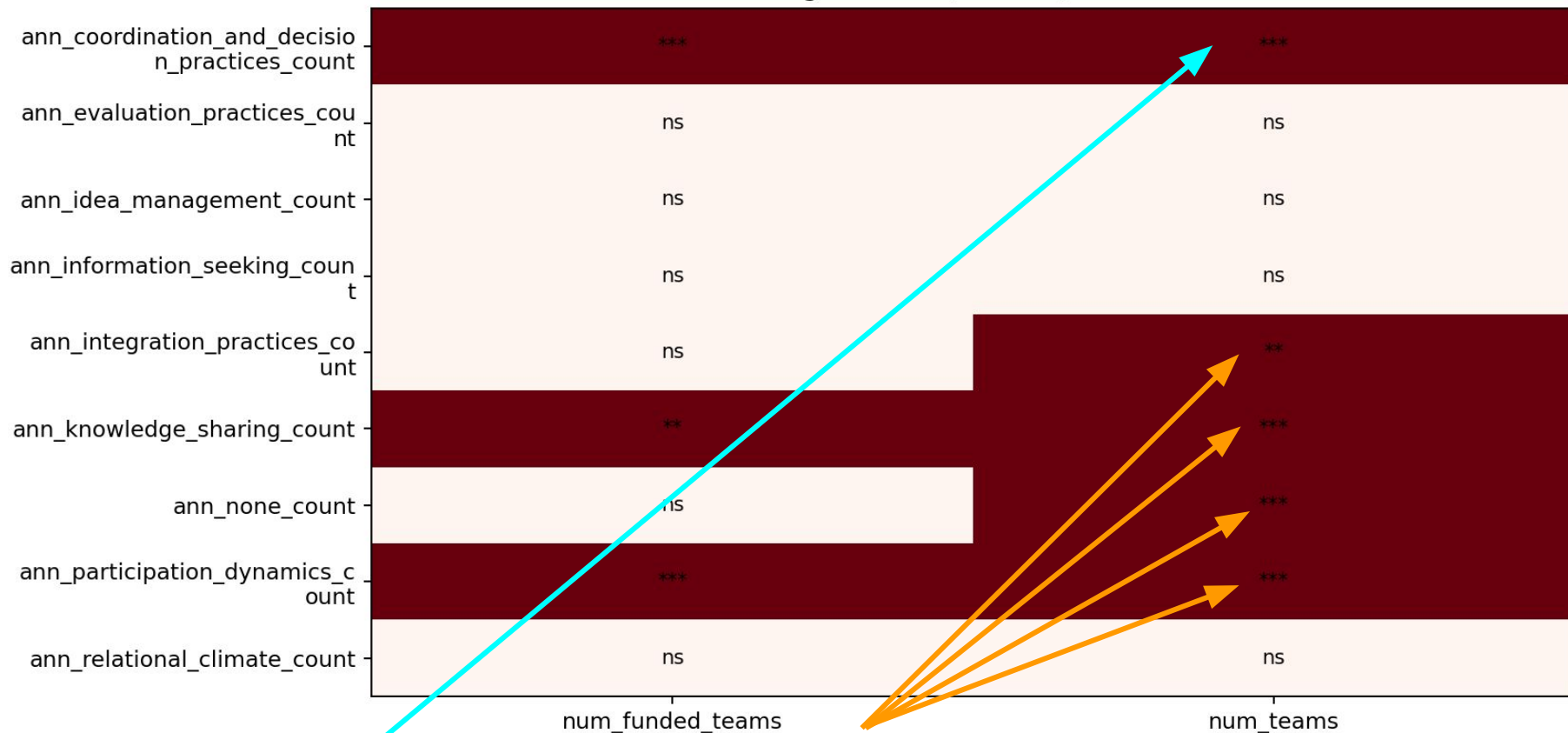
Participation dynamics and decision-related behaviors dominate, signaling facilitation and convergence toward decisions.

Significance ($\alpha=0.05$) — Middle

ann_coordination_and_decision_practices_count	***	***
ann_evaluation_practices_count	ns	ns
ann_idea_management_count	ns	ns
ann_information_seeking_count	ns	ns
ann_integration_practices_count	ns	ns
ann_knowledge_sharing_count	**	***
ann_none_count	ns	ns
ann_participation_dynamics_count	**	**
ann_relational_climate_count	ns	ns
	num_funded_teams	num_teams

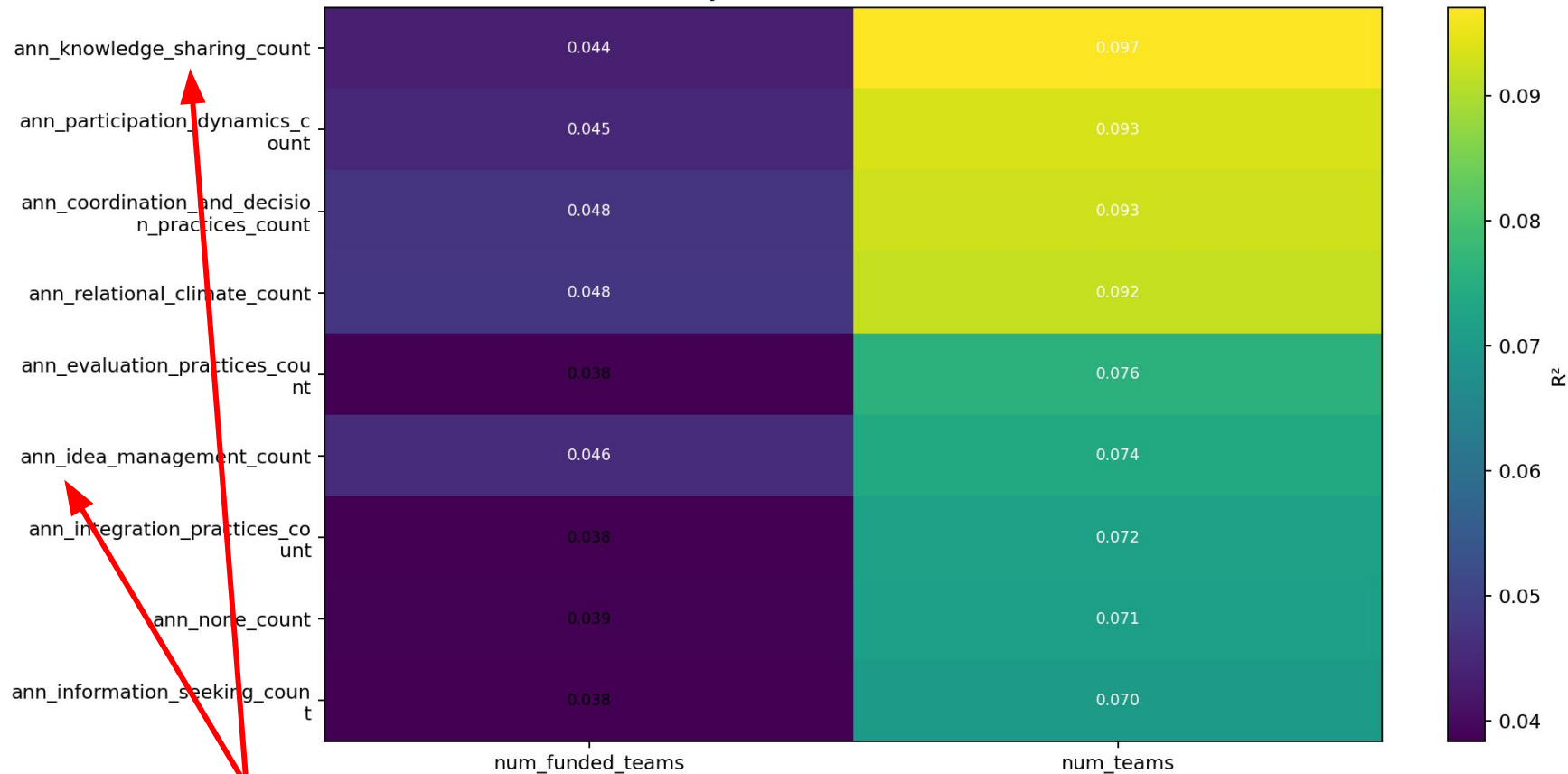
Mid-phase significance highlights the transition from idea exchange (early) to structured coordination and participation (later).

Significance ($\alpha=0.05$) — Last

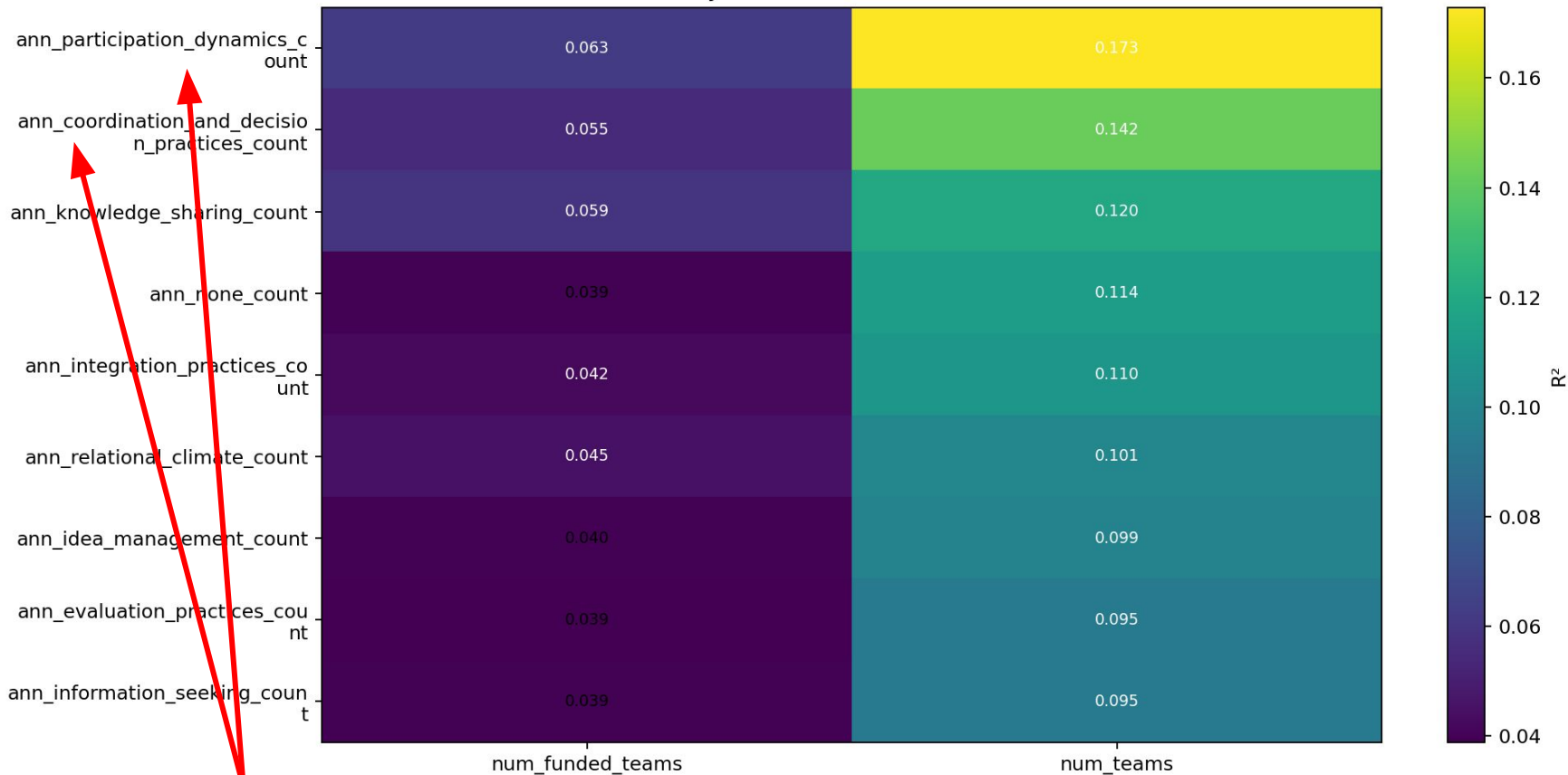


Decision-oriented coordination behaviors remain strong

Late-phase collaboration characterized by active participation and integrated problem-solving

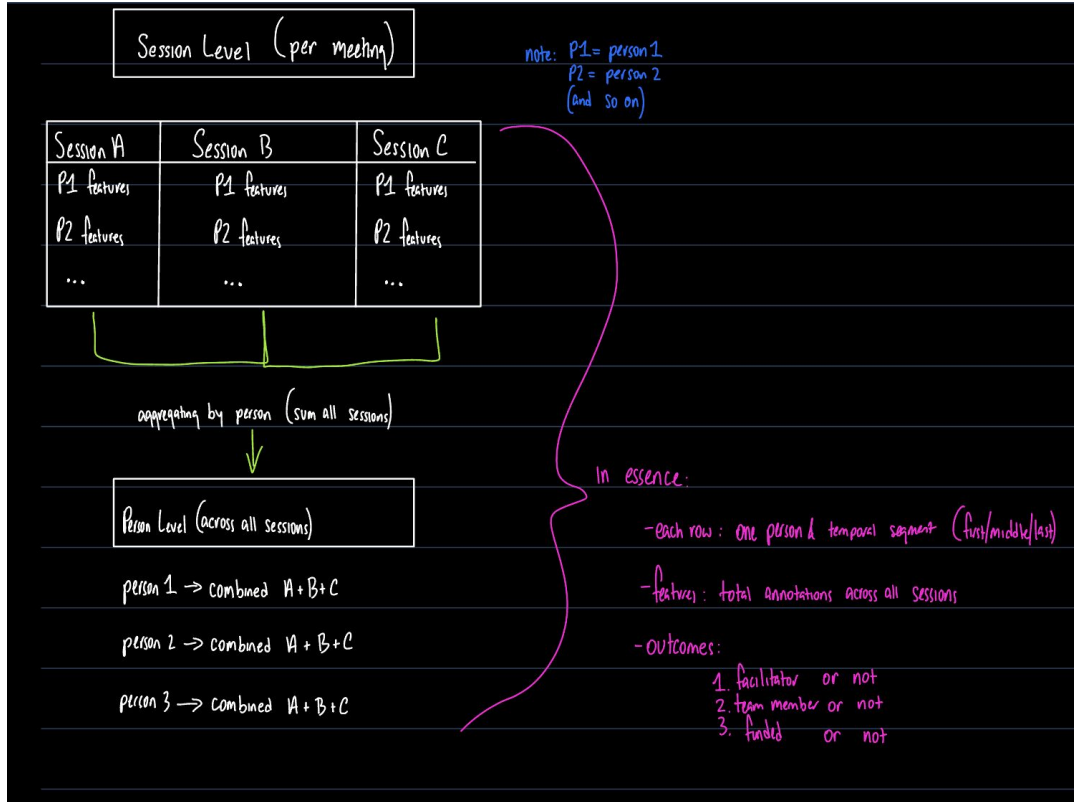
R² by Feature × Outcome — First

Idea sharing and knowledge exchange explain modest variance in outcomes ($R^2 \approx 0.04\text{--}0.09$)

R² by Feature × Outcome — Last

Coordination and participation codes explain the largest share of variance; showing structured collaboration

Answering Slide 101 (next several slides)



process of aggregating
across persons

Feature Glossary (person-level, temporal)

For the features:

- Knowledge Sharing, Evaluation Practices, Idea Management, Coordination & Decision Practices, Information Seeking, Relational Climate, Participation Dynamics, Integration Practices

Aggregation:

- Computed at the individual level (person-level), summed across all sessions & conferences
- Each person's behavior is summarized across sessions for each part of the meeting - first, middle, last.
 - By aggregating across sessions & conferences, this handles outliers.

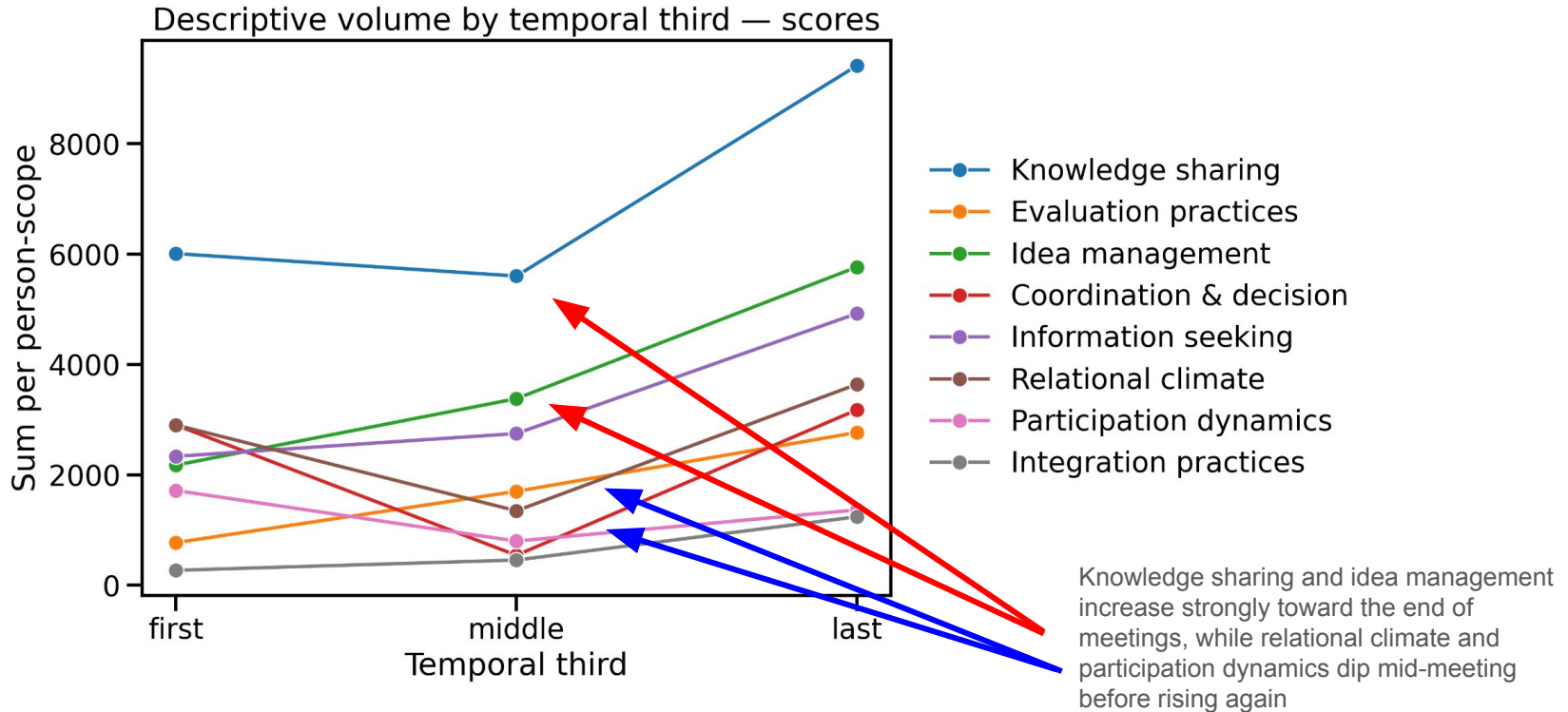
Models Applied:

- Used in OLS, Logistic, and Prediction regressions (+ Negative Binomial for zero-inflated outcomes)

Control Variables:

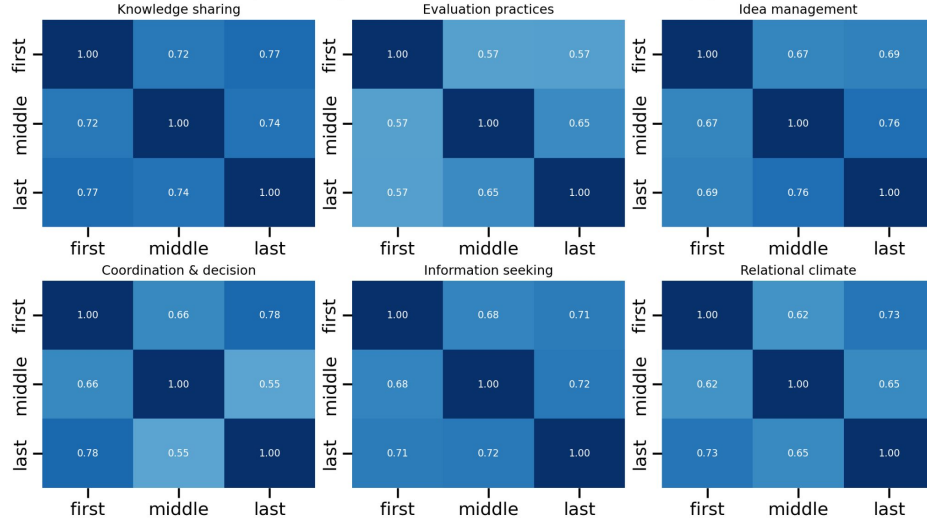
- Speaking durations (seconds) ... accounts for talkativeness
- Conference fixed effects ... isolates within-conference behavioral differences

Temporal Trends (Beginning, Middle, End) (person-level)

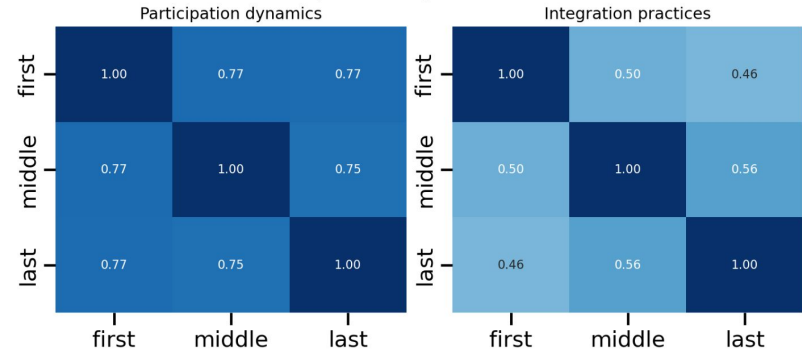


Temporal Correlations (Behaviors b/w One Another)

Within-person temporal correlations (first/middle/last) — scores (page 1)



Within-person temporal correlations (first/middle/last) — scores (page 2)

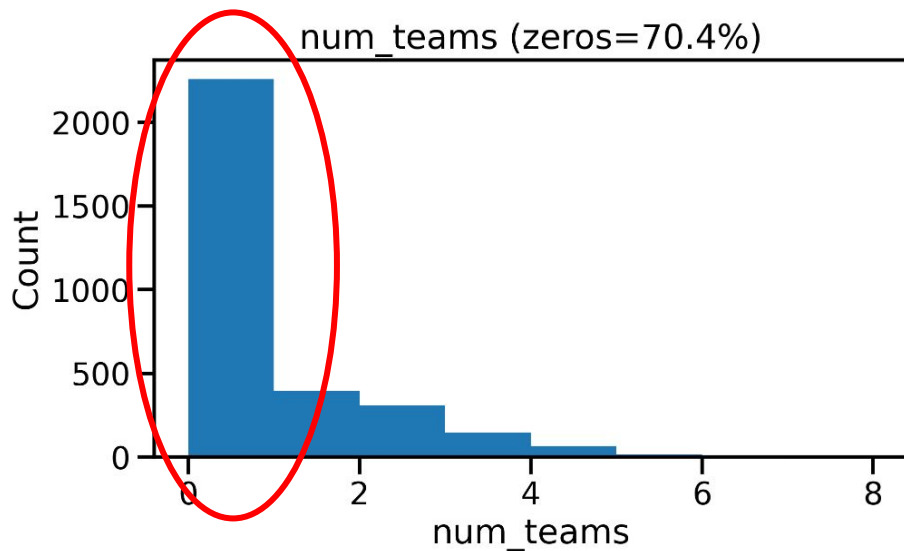


Knowledge Sharing and Idea Management remain consistent between each of the temporal thirds.

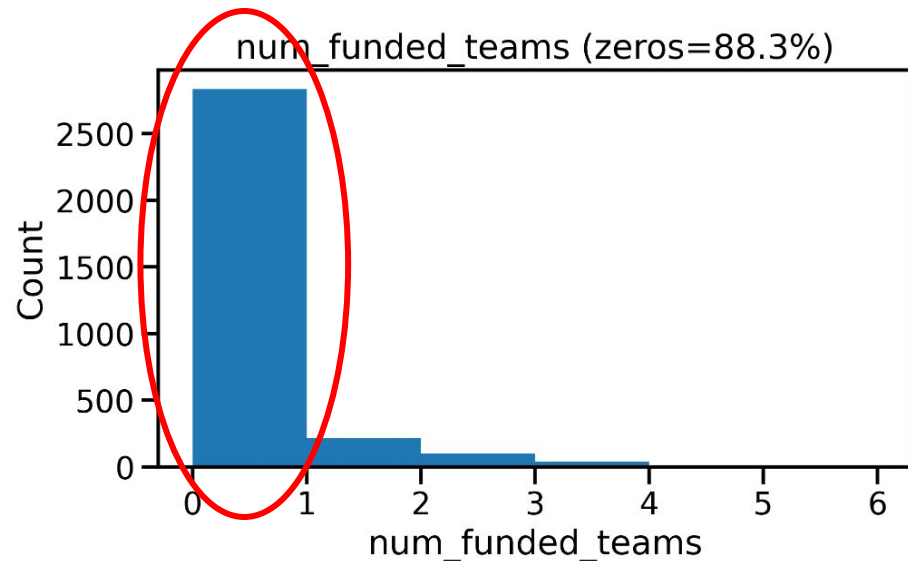
- Testing whether individuals' behaviors are stable over time - i.e., whether people who do more of something early continue to do so later on.
 - (This examines within-person continuity before testing outcomes tied to specific roles - facilitator, team member, or funded team.)

Outcome Distributions

Outcome distributions (check for zero-inflation)

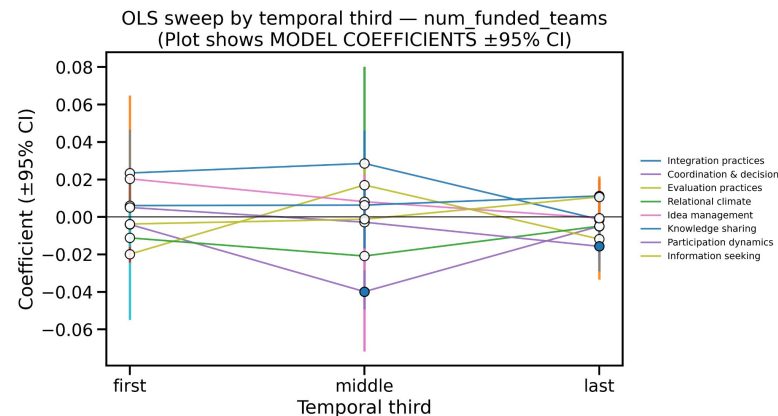
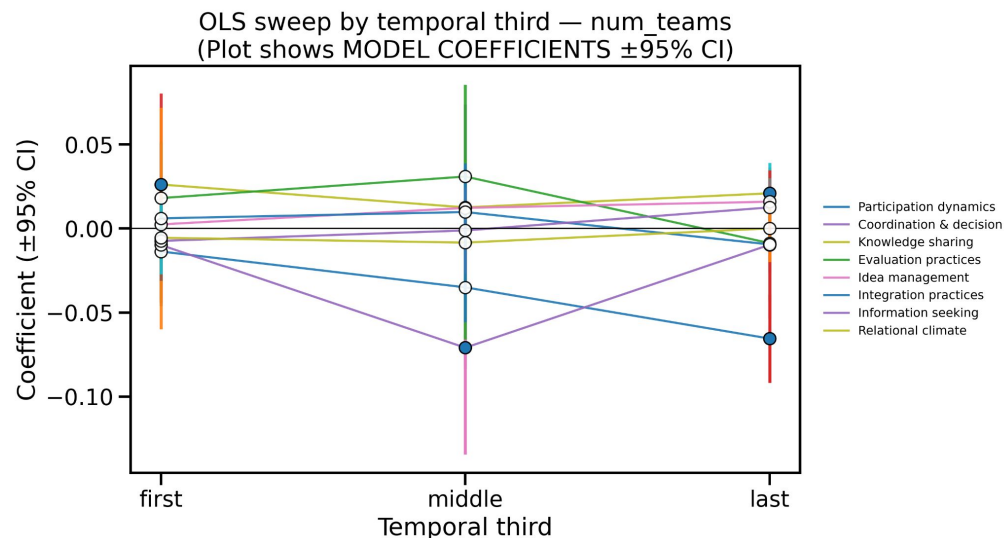


Most people aren't in any team or funded team, hence a lot of zeroes. This motivates a negative binomial in addition to OLS



Additional note: facilitator is a binary variable (0 = non-facilitator, 1 = facilitator), so this would be examined separately using logistic regression rather than a count distribution

Regression Results (OLS + Logistic Regression) + Negative Binomial



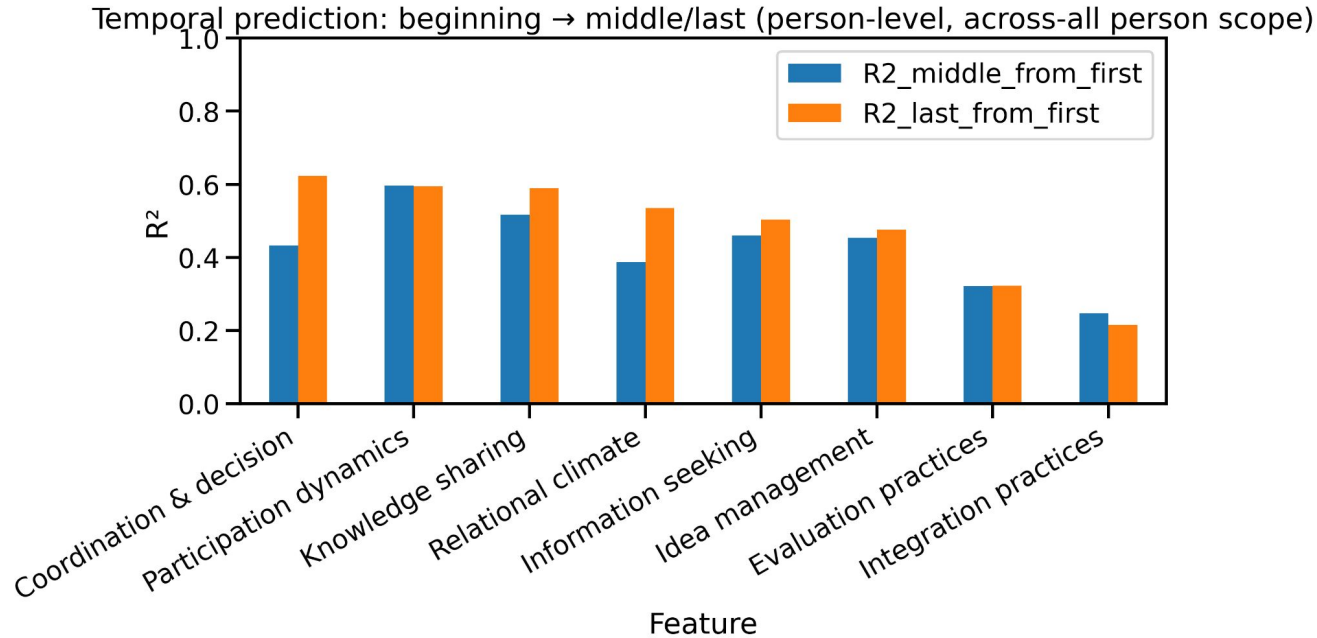
Negative / positive coefficients indicate fewer or more teams (or funded teams).

Behavioral effects tend to intensify during the middle third of meetings... timing playing role in how behaviors turn into outcomes.

Controls: speaking duration (sec), conference fixed effects

Negative binomial: used to check num_funded_teams due to zero inflation ($\approx 88\%$) -> checked with OLS results

Temporal Prediction (First -> Middle / Last)

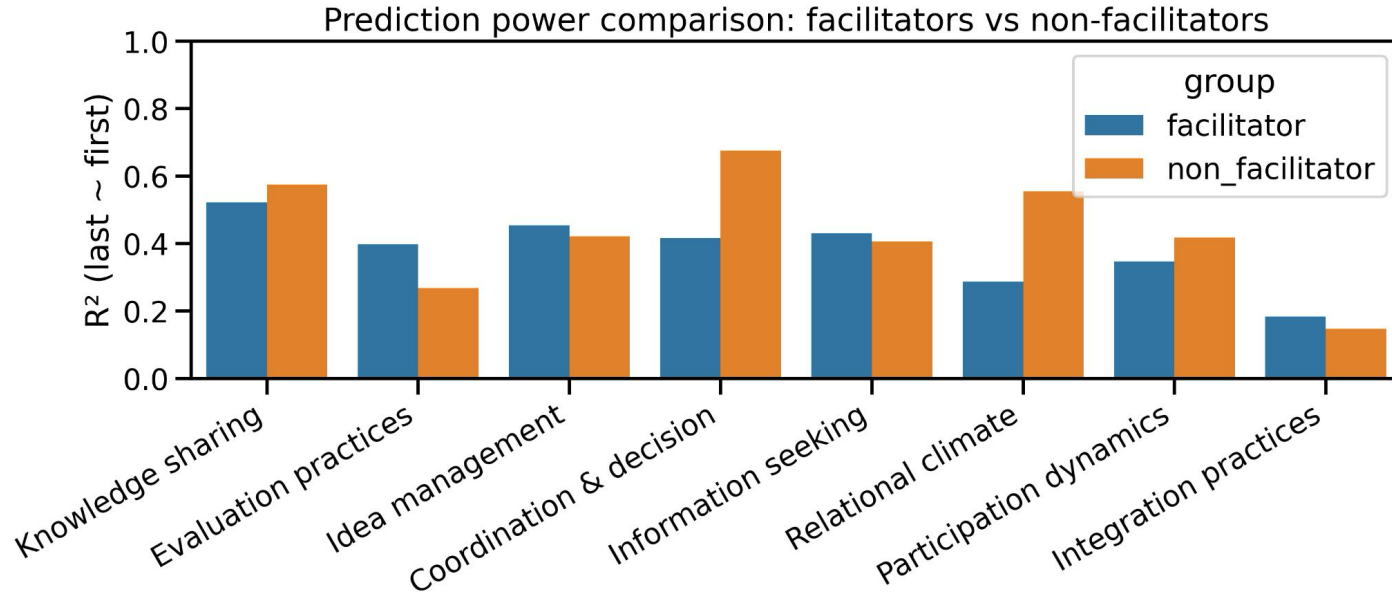


Early behaviors (first 1/3 segment) strongly predict later ones showing that individuals' behavioral patterns remain consistent through meetings

Blue: comparing first segment to middle segment

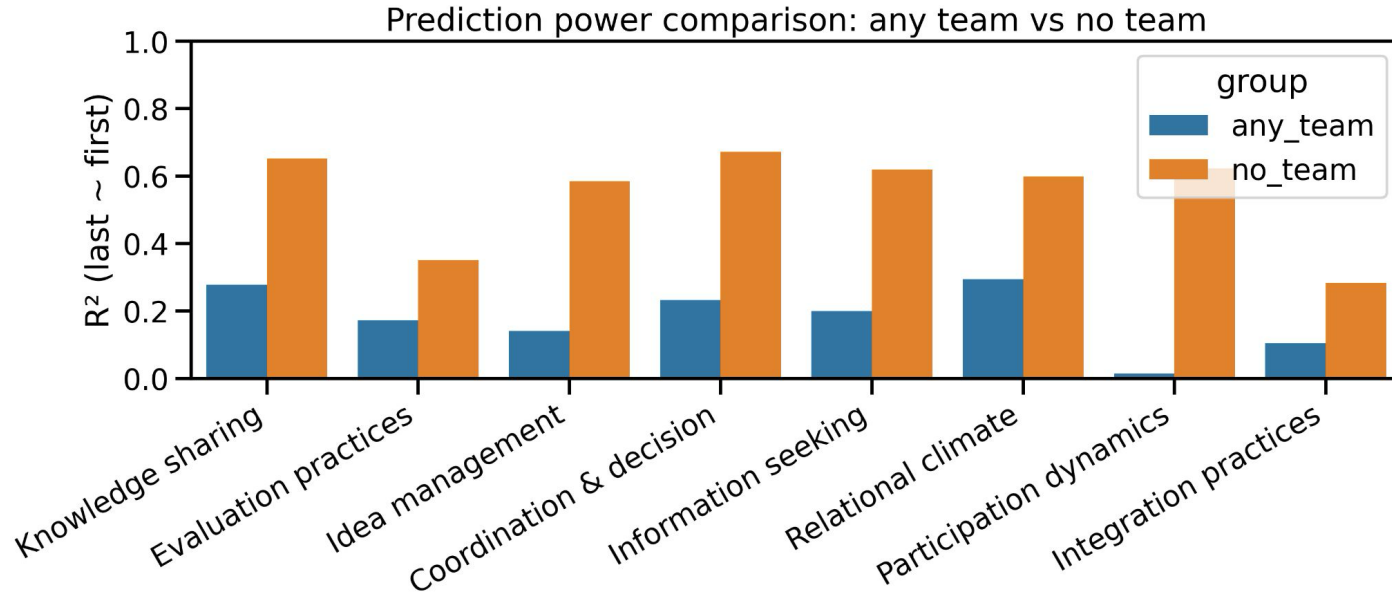
Orange: comparing first segment to last segment

Facilitator vs. Non-Facilitator Role



Facilitators show slightly stronger within-person prediction for relational and coordination behaviors... suggesting more consistent behavioral style across meetings while guiding structure

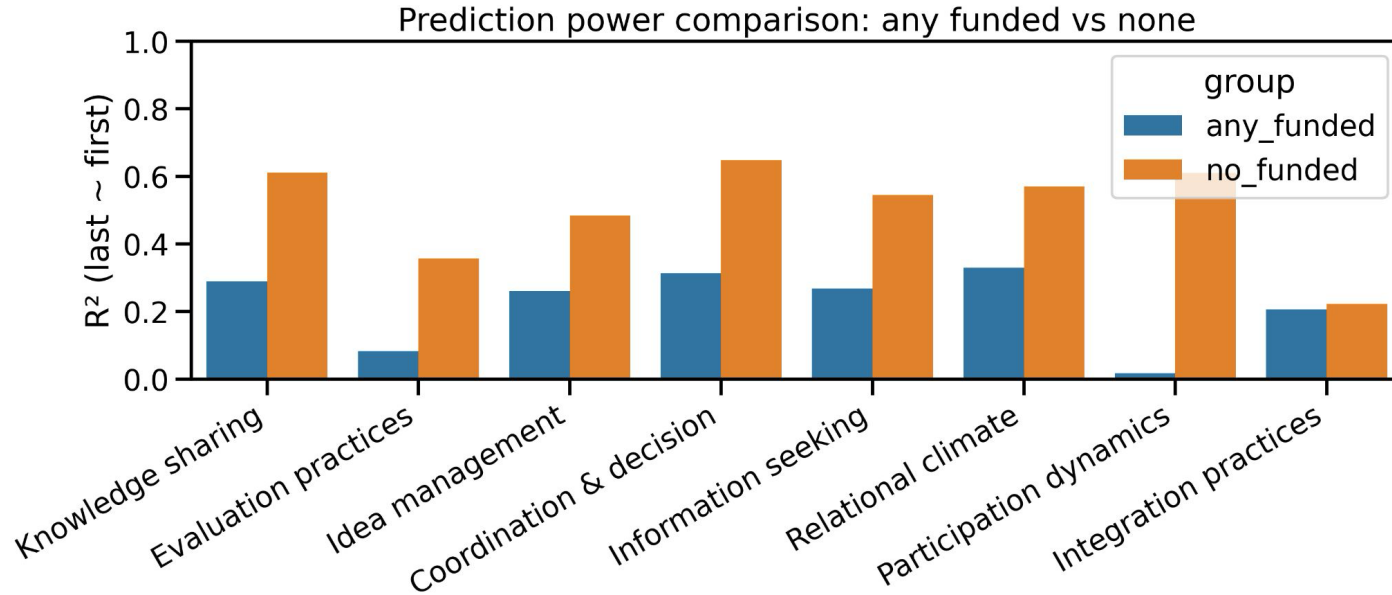
On-Team vs. No-Team



People not on teams (orange) show higher behavioral consistency (thus, higher R^2).

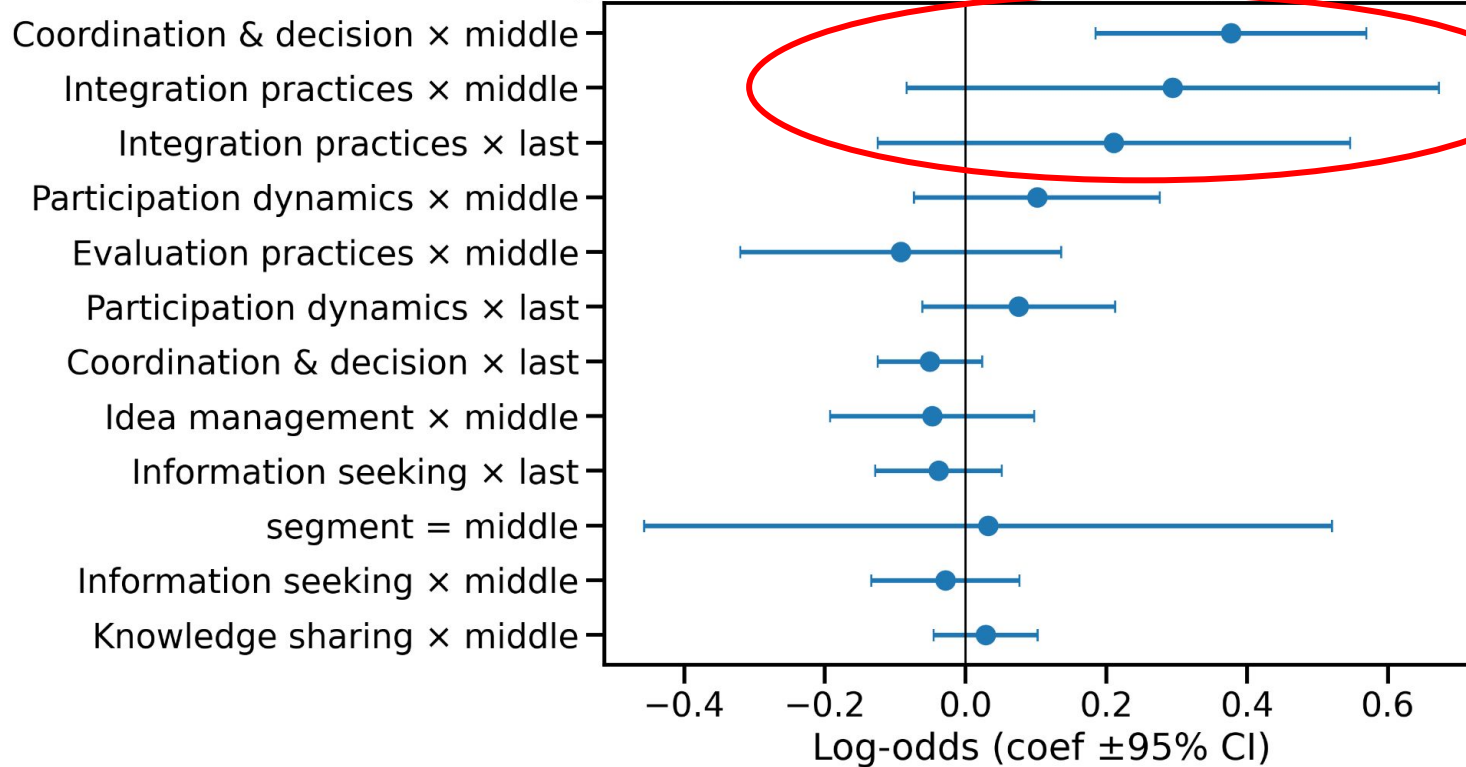
Team members (blue) vary more possibly due to shifting coordination roles/behavioral adjustment & responsiveness over time

Funded Team vs. Non-funded Team



People not linked to funded teams show stronger behavioral stability across meeting (higher R^2)
Funded participants vary more - possibly due to adapting behaviors / focus over time

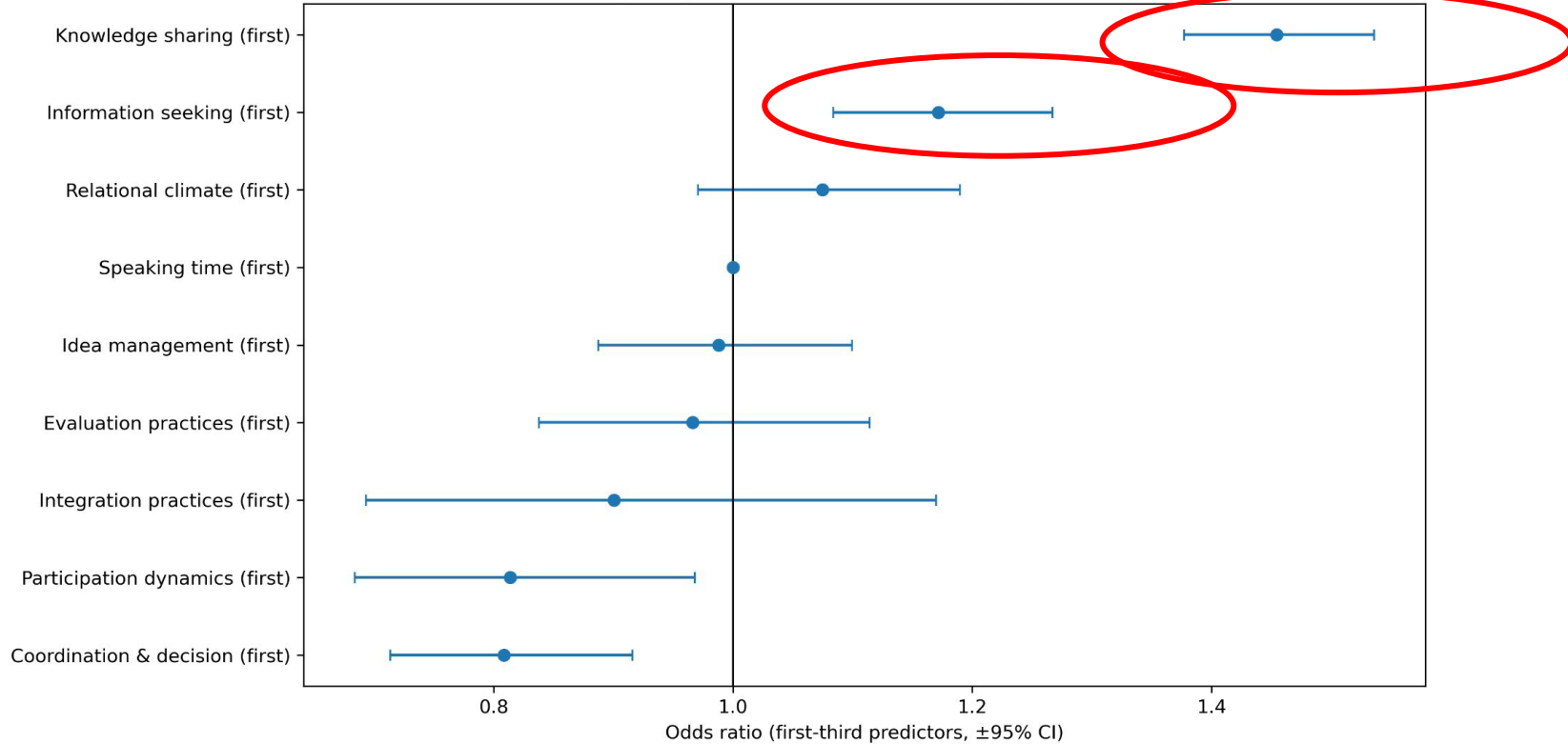
Logistic regression — facilitator × segment interactions



Facilitators show stronger Coordination & Decision and Integration Practices during the middle of the meetings - the part where structure and alignment matter most.

Facilitators taper in Integration Practices by the final third, suggesting facilitators step back as discussion emerges (although this pattern shows high uncertainty with a wider CI).

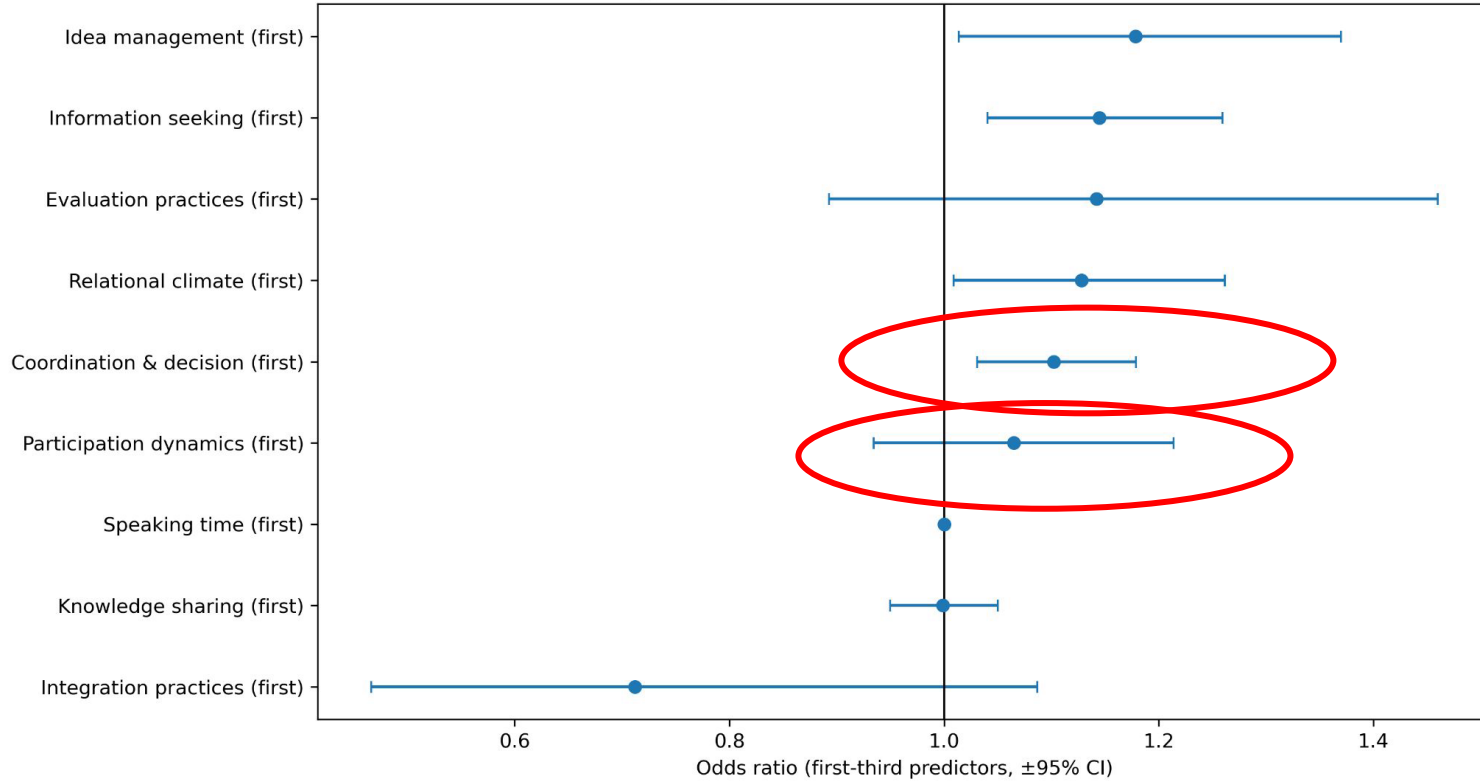
Beginning (first-third) behaviors → Any team (1/0)



High Knowledge Sharing and Information Seeking in the first $\frac{1}{3}$ predict a greater likelihood of being on a team

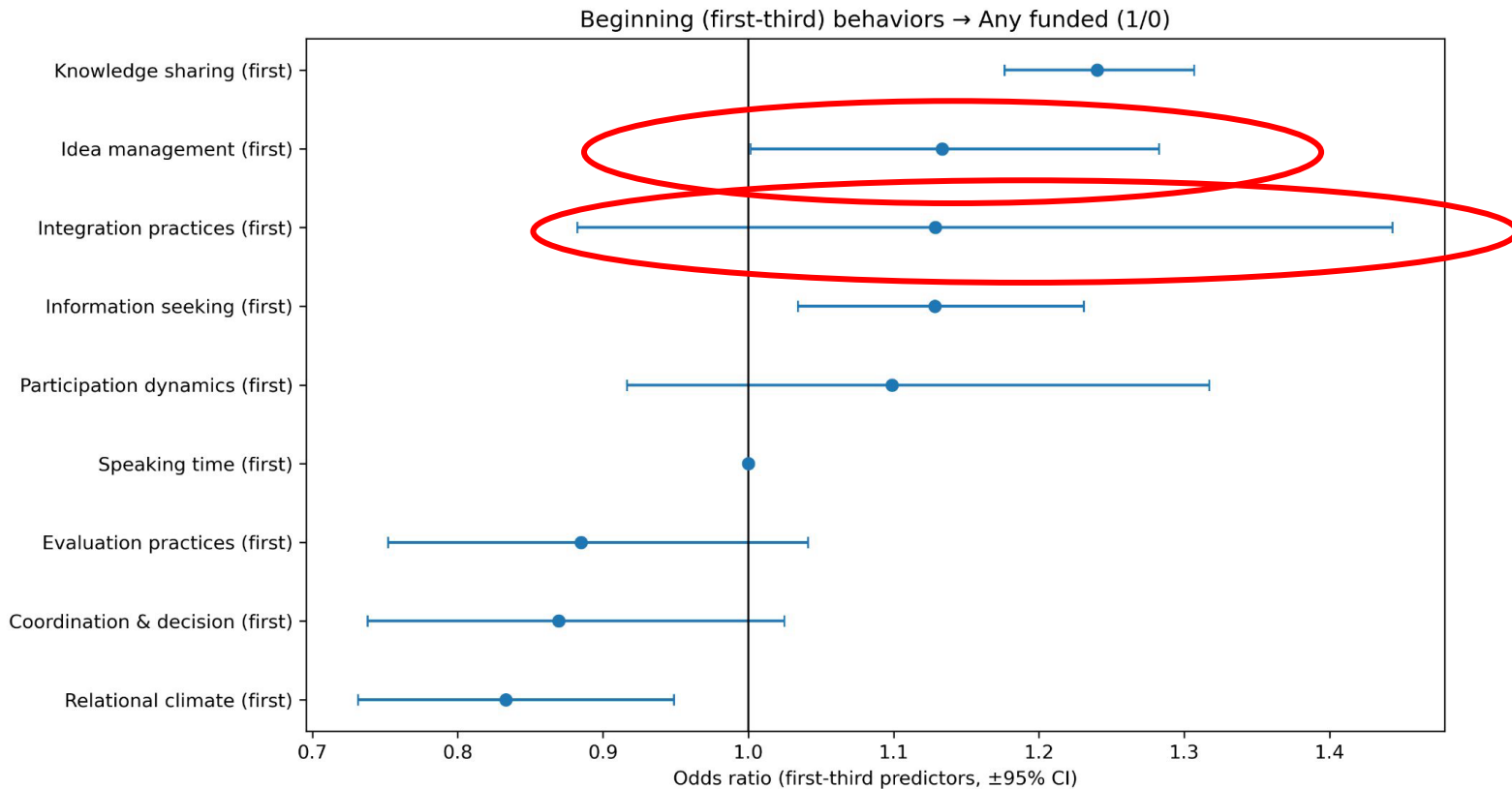
Early information exchange may signal collaborative intent or awareness of others' contributions

Beginning (first-third) behaviors → Facilitator (1/0)



Individuals show higher Coordination & Decision and Participation Dynamics during the first third of meetings are more likely to act as facilitators.

Early structure-setting and engagement behaviors appear to differentiate facilitators from others before the meeting processes



Individuals demonstrating more Idea Management and Integration Practices early in meetings are more likely to be in funded teams

This suggests that early idea organization and synthesis behaviors may contribute to success or recognition in funding outcomes

Findings about Early Behaviors Predicting Later Roles

Non-facilitators/non-team/non-funded participants show higher R^2 , more stable, less adaptive behaviors

Facilitators, team members, and funded participants show lower R^2 , greater behavioral flexibility throughout the three meeting stages (beginning, middle, end)

For the features in the first $\frac{1}{3}$ of meetings:

- Early Coordination & Participation ... more likely facilitator
- Early Knowledge Sharing & Information Seeking ... more likely team members
- Early Idea Management & Integration ... more likely funded teams

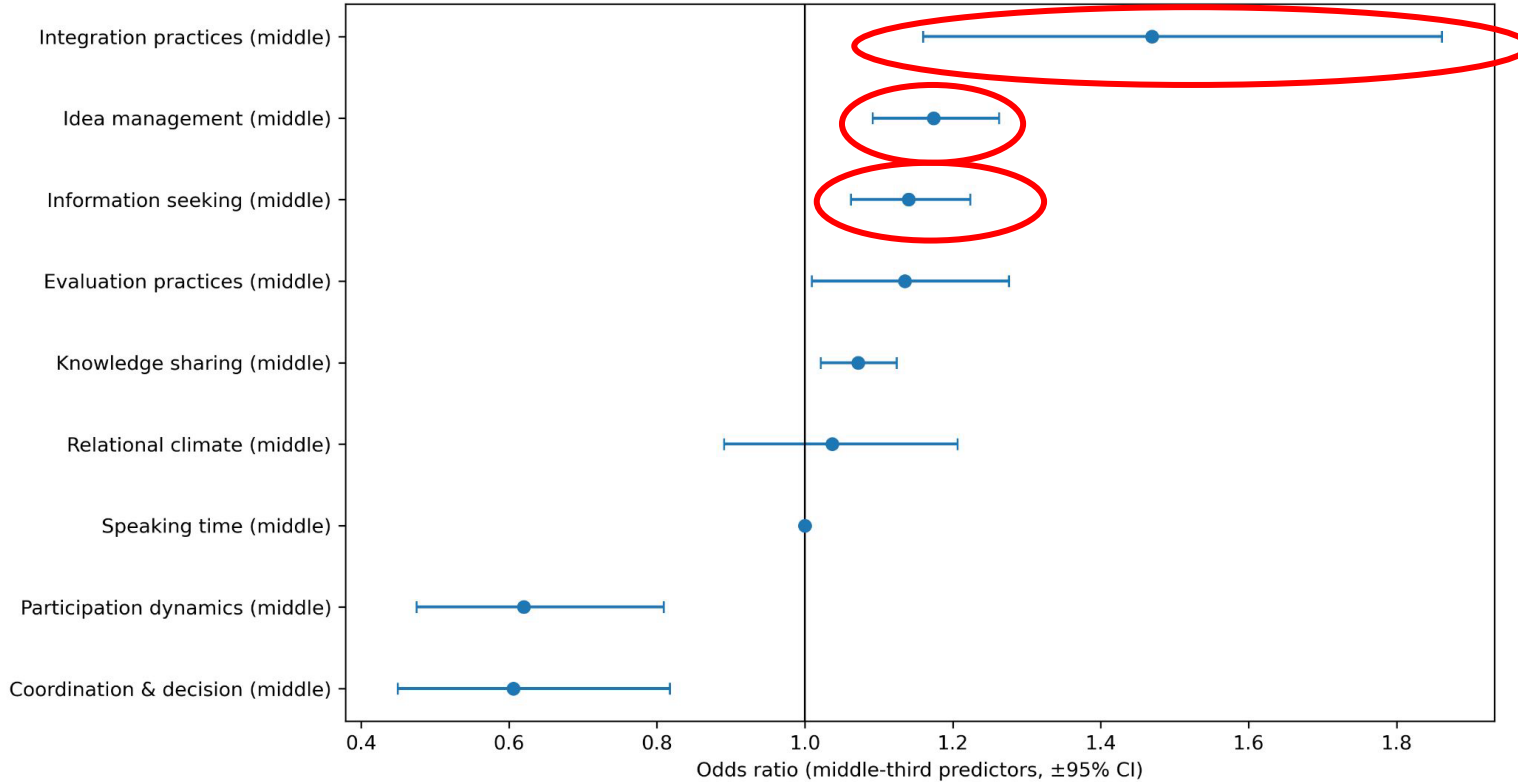
Would it be ideal to start predicting middle features and seeing if they have impact on end?

Predicting Middle Features & Seeing If They Have Effect on Final $\frac{1}{3}$ segment of session for predicting roles

Done naturally after figuring out that the first temporal section has impact on both middle and end section

- → so then does the middle section have “direct impact” on the last temporal third?

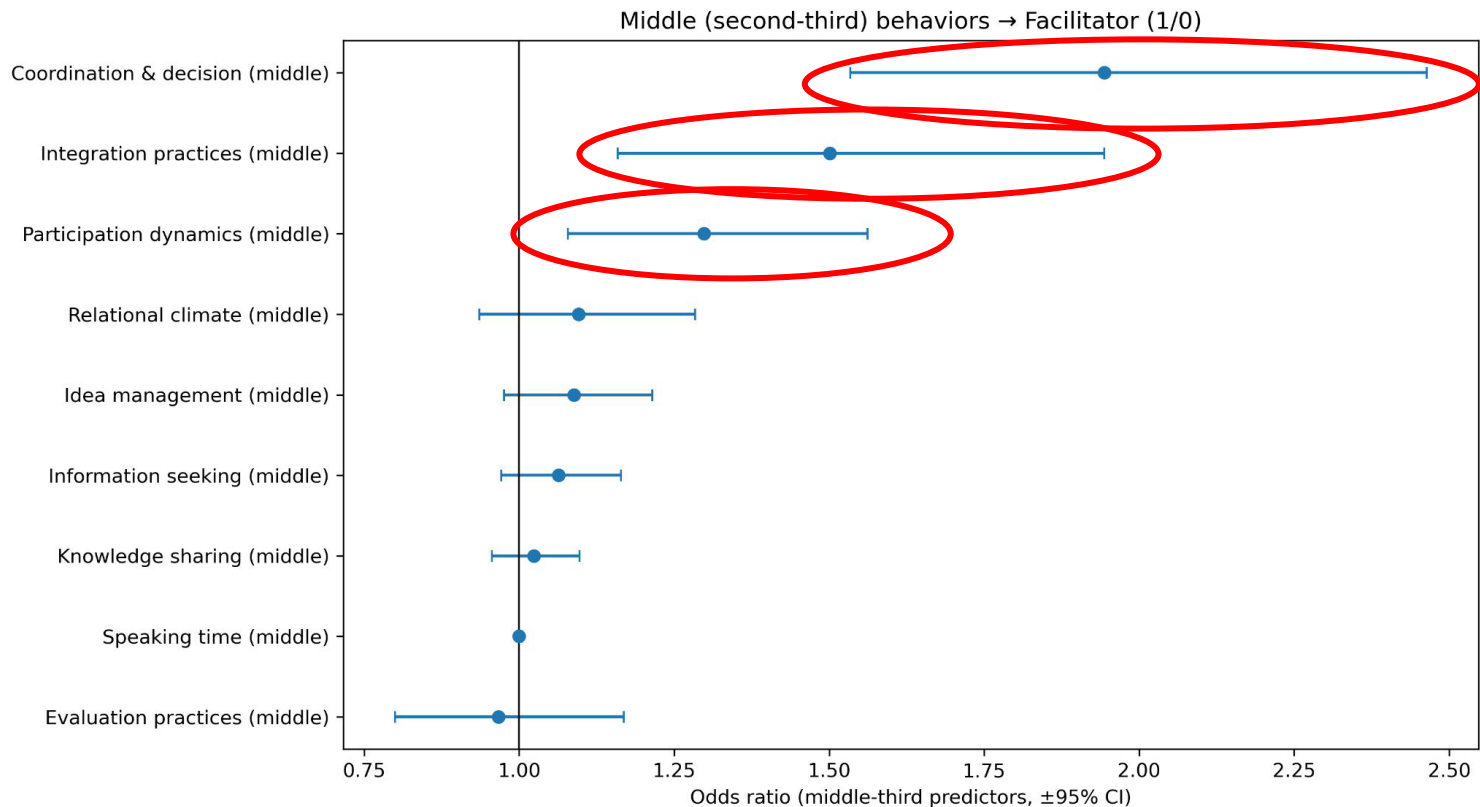
Middle (second-third) behaviors → Any funded (1/0)



Teams with funding have an uptick mid-meeting; connecting insights/narrative, cross member contributions, etc. (Integration Practices)

Show the ability to refine/iterate ideas efficiently (small CI) (Idea Management)

Show the ability to check assumptions and filling conceptual gaps (small CI) (Information Seeking)

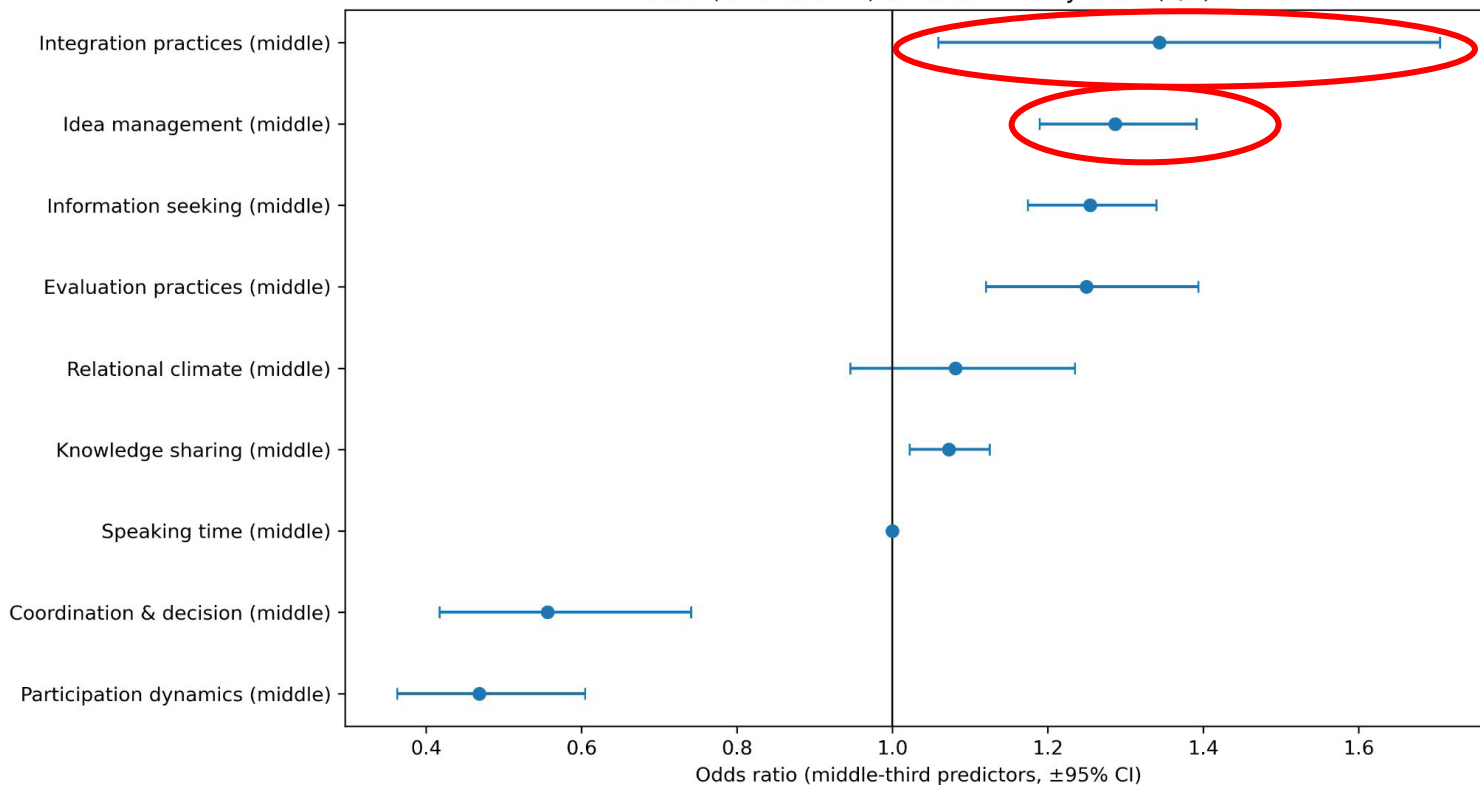


Facilitators structuring how group converges, guides tasks during second-third segment (Coordination & Decision Practices)

Facilitators synthesizing perspectives and others' contributions mid-discussion (Integration Practices)

Facilitators shape the flow of who speaks and when (Participation Dynamics)

Middle (second-third) behaviors → Any team (1/0)



Those who integrate ideas into the middle of meeting more likely to belong to a team (Integration Practices)

Those who integrate ideas mid-meeting and help ideas evolve (Idea Management; Information Seeking) more likely to belong on a team

Those who maintain interpersonal trust during the middle-segment with others (Relational Climate)

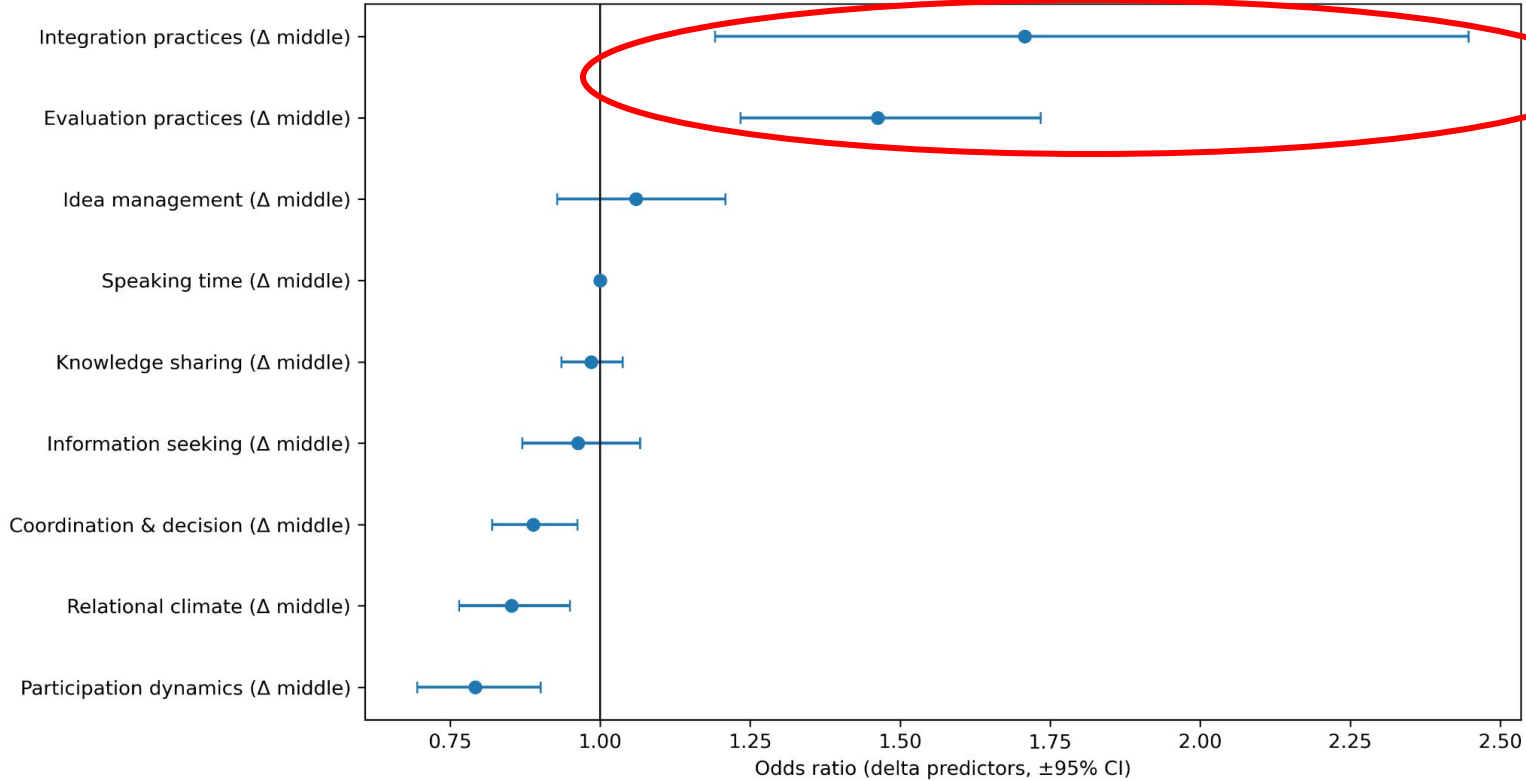
Eventually looking at the overall perspective (person-level)

How do behavioral changes across the meeting (first → middle → last) differentiate from people who achieve meaningful outcomes (facilitation, teamwork, funding) from those who remain static?

Incorporating changes in delta for role & segment-third ...

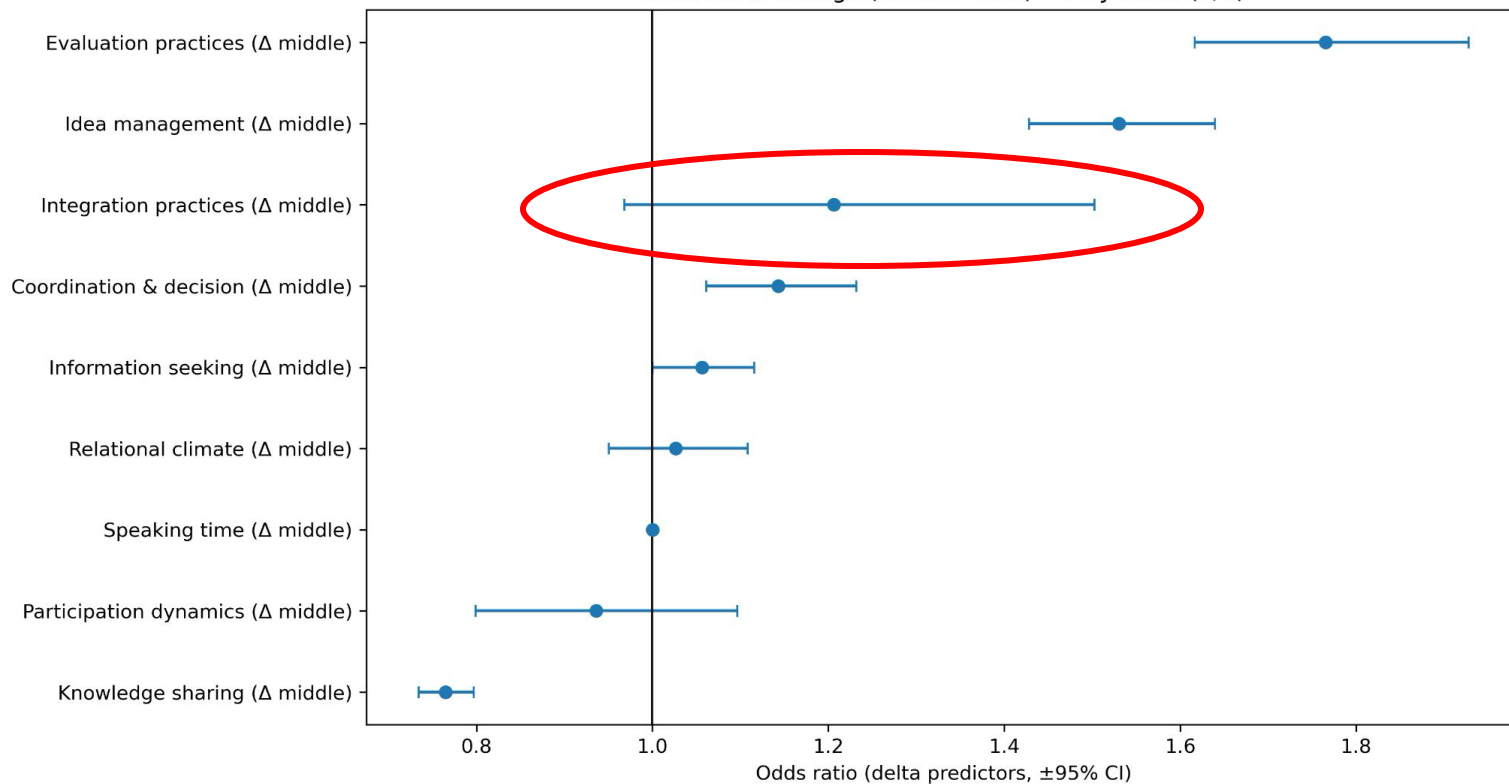
Looking at the changes between first $\frac{1}{3}$ segment and middle $\frac{1}{3}$ segment and middle $\frac{1}{3}$ segment and last $\frac{1}{3}$ segment

Behavioral change (middle–first) → Facilitator (1/0)



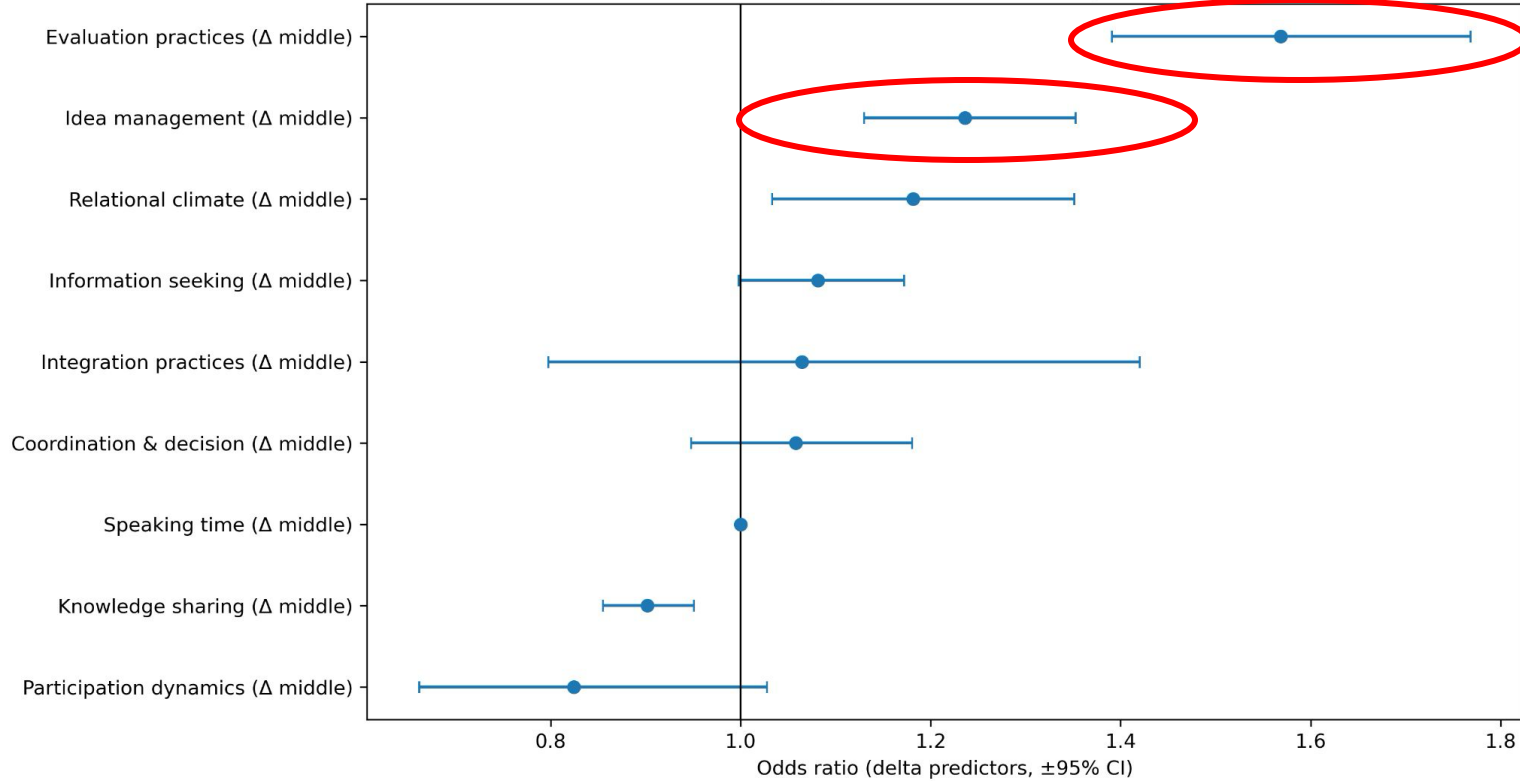
Facilitators show increasing activity in Integration Practices and Evaluation Practices mid-meeting

Behavioral change (middle–first) → Any team (1/0)



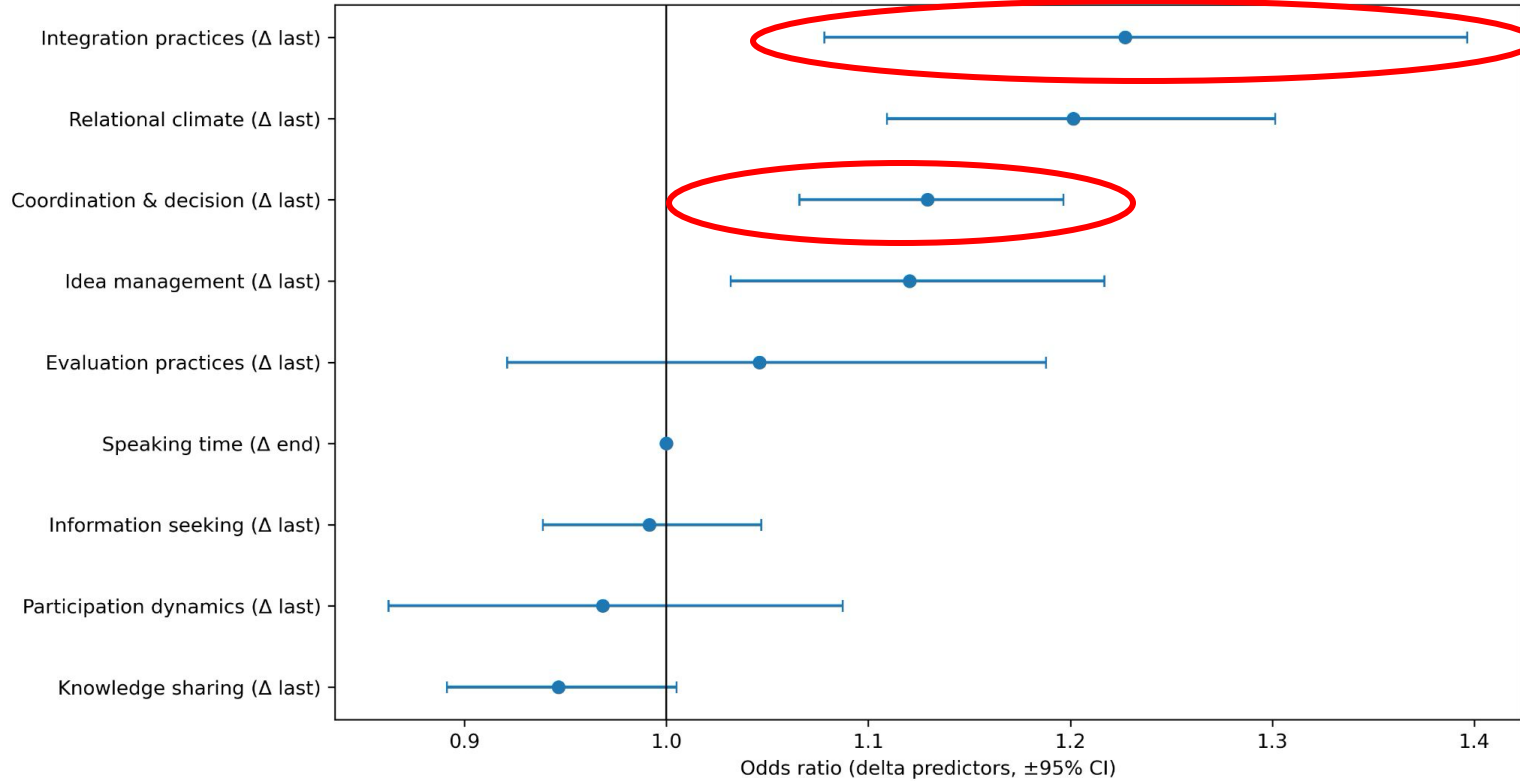
team members increase their Idea Management and Integration Practices & Relational Climates

Behavioral change (middle–first) → Any funded (1/0)



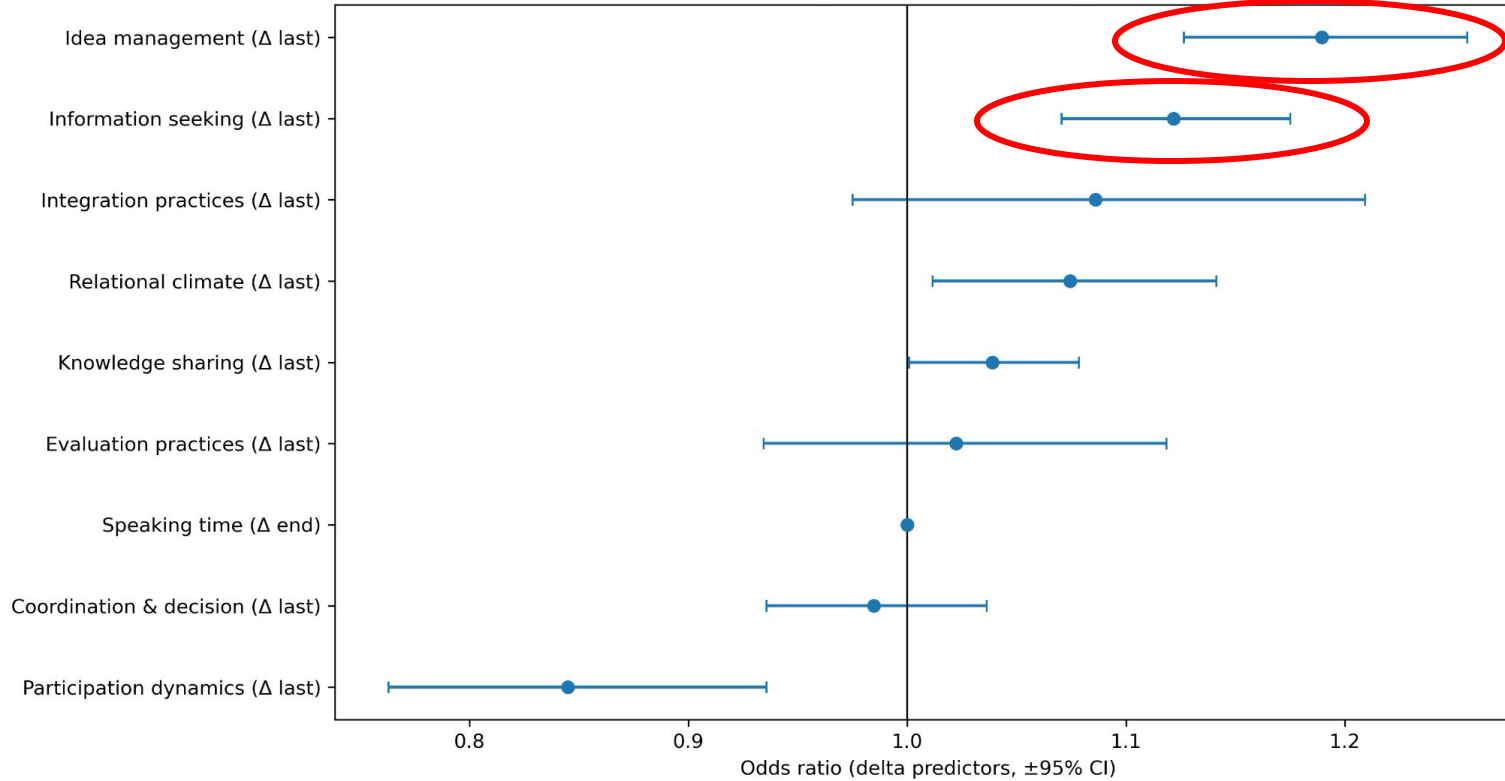
Funded teams show gains in Evaluation Practices and Idea Management → due to outcome-oriented discussion by the middle of the meeting

Behavioral change (last-middle) → Facilitator (1/0)

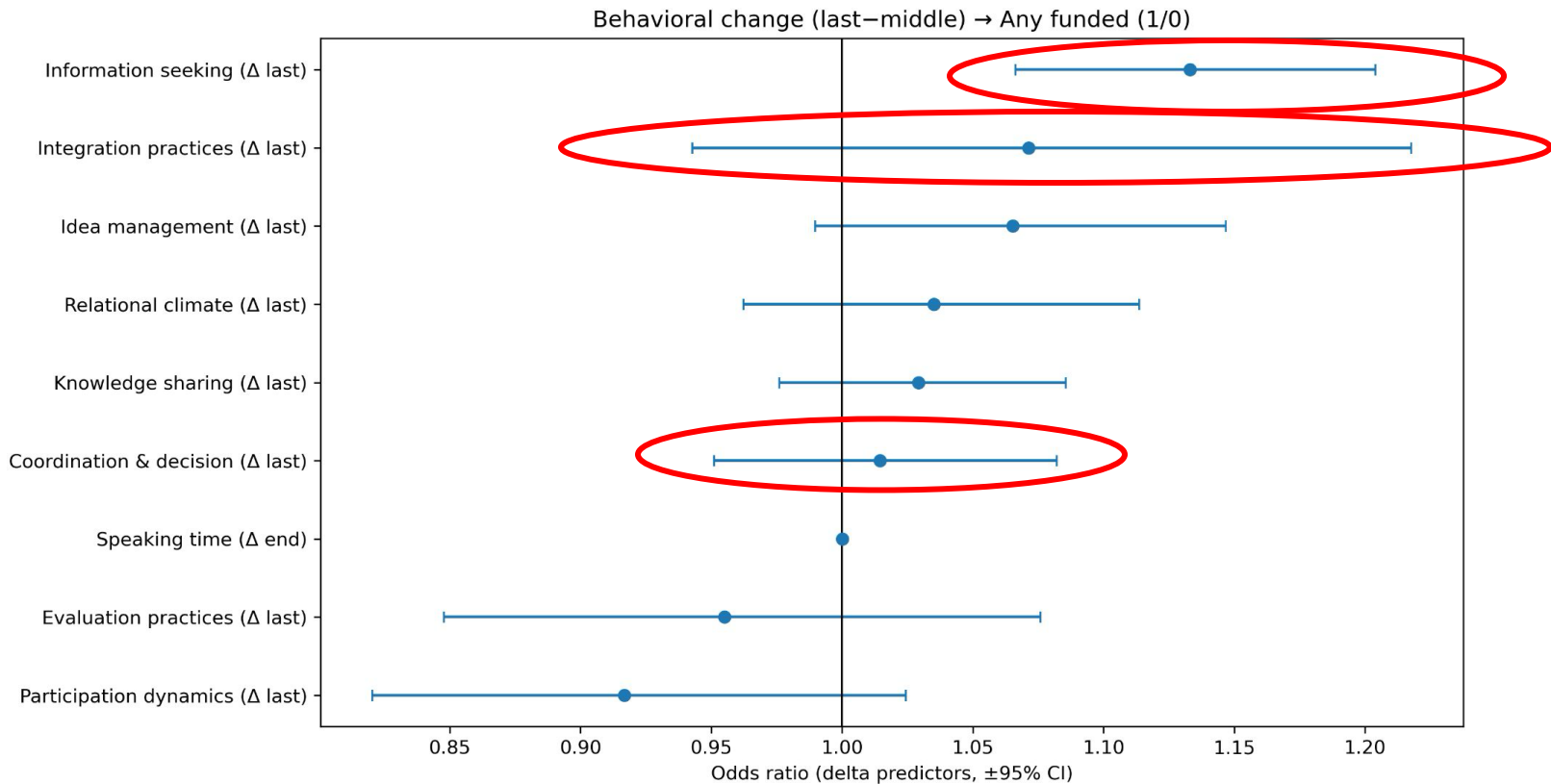


Facilitators maintain strong Integration Practices and Coordination & Decision Processes going from middle-segment onto the final-segment → keeping structured decisions and convergence points

Behavioral change (last–middle) → Any team (1/0)



Teams' final segment increases in the Idea Management and Information Seeking, show reflection & gap-checking behaviors, wrapping it up properly before session comes to a closure.



Funded teams have the most balanced pattern out of the three roles (is-facilitator, on-team, is-funded). From the middle to the end, they maintain the focus and coordination achieved earlier in the meeting instead of changing the behavior drastically → “riding the wave to the end.”

Findings on Person-level & Role Difference

Facilitators:

- Show the strongest increases in Coordination & Decision and Integration Practices by the middle-segment of a session.
- These behaviors rise in the middle-segment of the session (where the structure is critical for end results) and then tapers by the last-segment
 - This shows that facilitators by their features are able to hold an adaptive structuring.

Team Members:

- From the early to the middle segments, this role strengthens in the Integration and Relational Climate.
 - Team member strengthen more so from the early point of the meeting to the middle part of the segment.
 - Their collaboration is mostly dynamic more so than static

Funded Teams:

- In the middle-segment to the end-segment, there is much Information Seeking and Integration
- This is the continuous learning and aligning through time, no early spikes

Transition Matrix on person-level ?